



FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

Institute for Factory Automation
and Production Systems



Friedrich-Alexander-Universität
Technische Fakultät

F9 - Mechanical components and assembly of electric machines

Lecture Production of electric motors and machines

Alexander Vogel, M.Sc.

The assembly of electric machines takes place at the very end of the process chain and therefore has a very close impact on customer's satisfaction.

In order to understand the issues involved in the production of electrical machines, essential knowledge in assembling and joining steps over the whole manufacturing chain are required.

After attending the lecture unit "Mechanical components and assembly of electric machines" you should

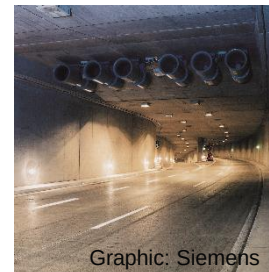
- know the procedure within the pre-assembly of the rotor,
- be informed about joining processes during stator pre-assembly,
- be familiar with required supporting processes
- and have got a full overview about all relevant final assembly steps and technics.



Graphic: Omron



Graphic: ZF Friedrichshafen



Graphic: Siemens



Graphic: Kuka

This unit covers the assembly processes used in the manufacture of electrical drives for components that do not directly affect electromagnetic energy conversion.

■ **Motivation and positioning of assembly within the process chain**

■ **Pre-assembly of the rotor shaft**

- Production and assembly of shaft and rotor
- Balancing the rotor shaft

■ **Pre-assembly of stator and housing**

- Production and joining process of stator and housing
- Mounting the bearing shield

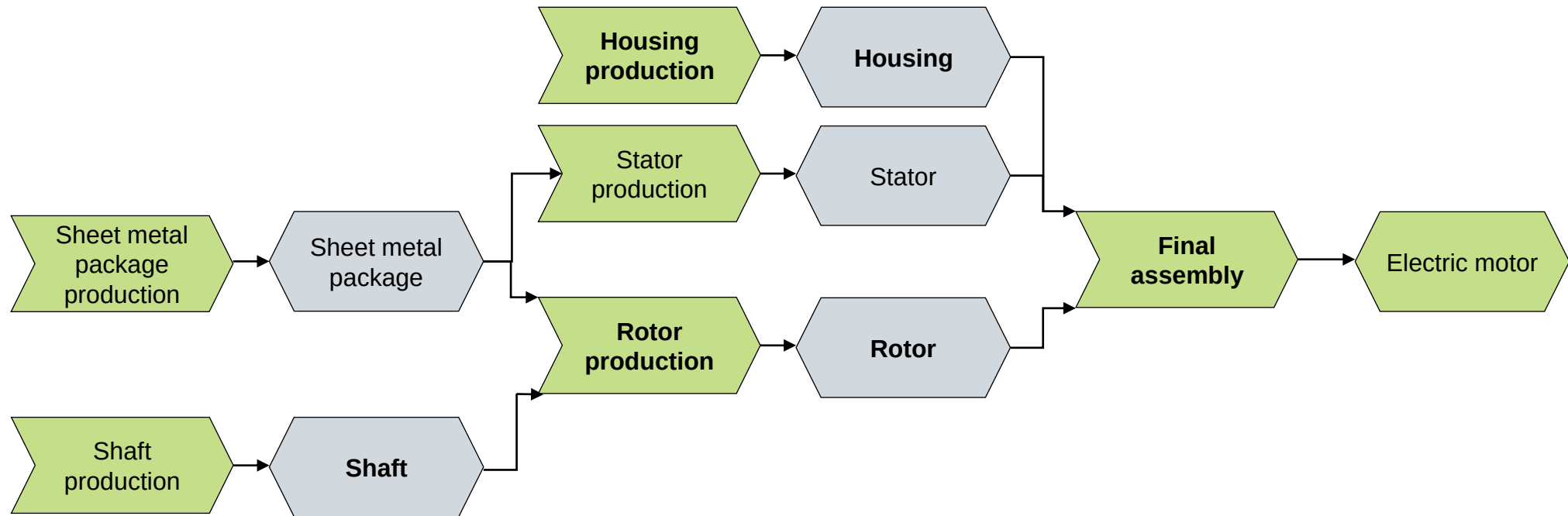
■ **Final assembly of the modules**

- Joining rotor shaft and stator
- Further motor electronics



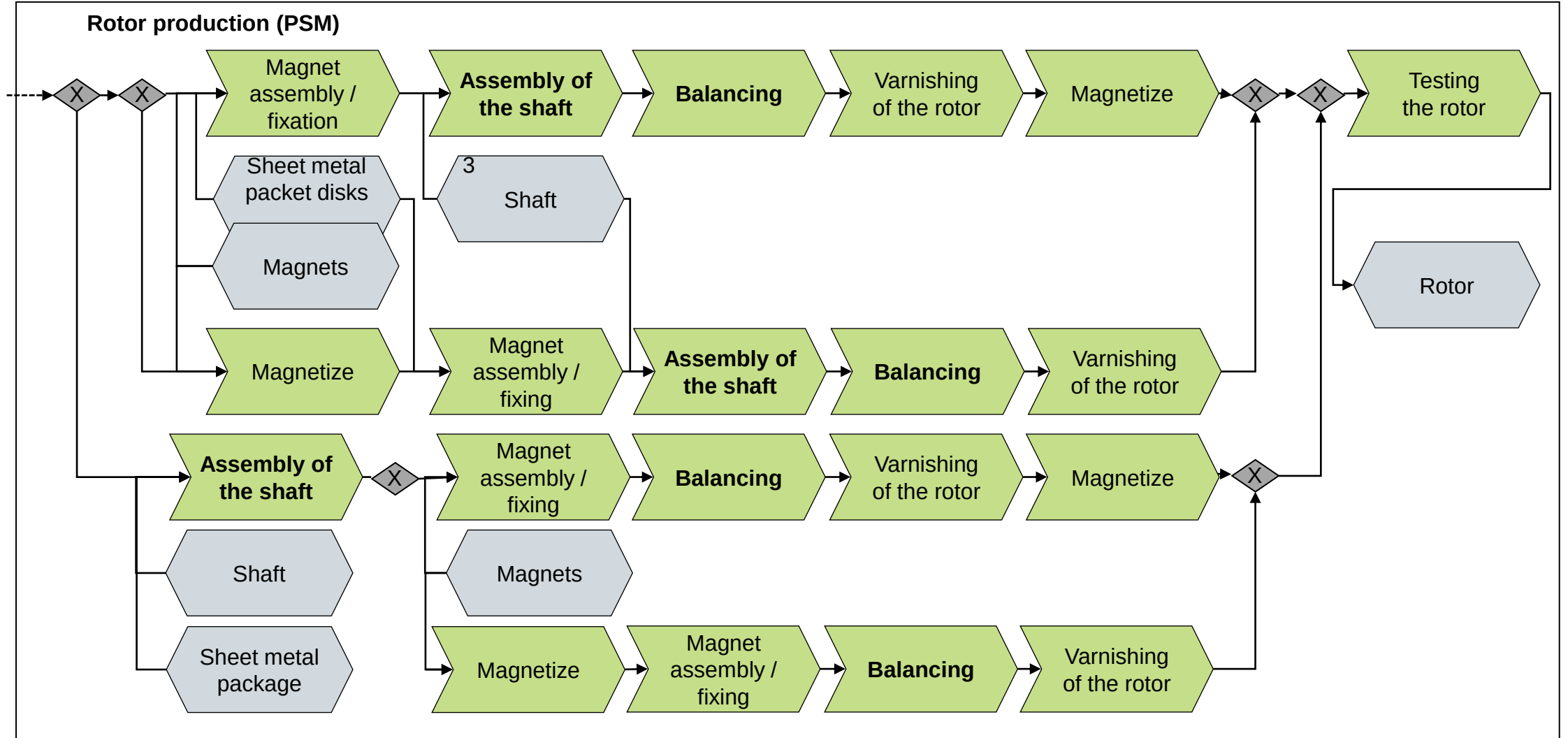
The process chain for manufacturing an electric motor comprises a large number of process steps and material elements.

Slide from Lec. 1



In the previous LUs, the processes and corresponding components relevant to the magnetic circuit were examined more closely.

Slide from Lec. 4

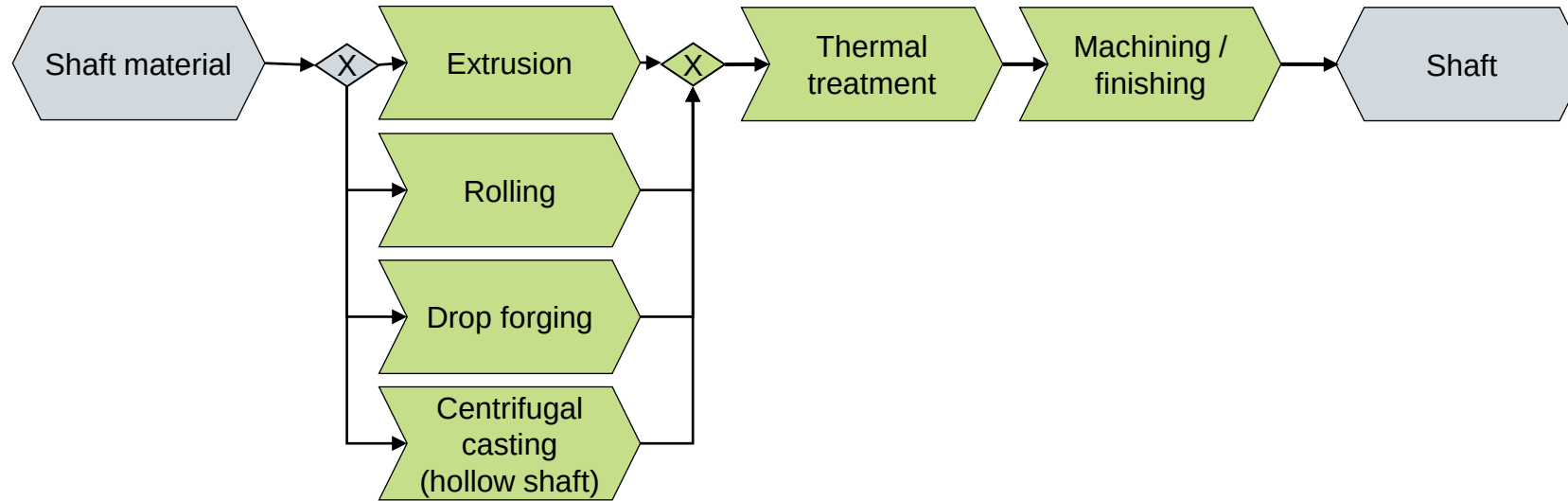


This unit covers the assembly processes used in the manufacture of electrical drives for components that do not directly affect electromagnetic energy conversion.

- Motivation and positioning of assembly within the process chain
- **Pre-assembly of the rotor shaft**
 - Production and assembly of shaft and rotor
 - Balancing the rotor shaft
- **Pre-assembly of stator and housing**
 - Production and joining process of stator and housing
 - Mounting the bearing shield
- **Final assembly of the modules**
 - Joining rotor shaft and stator
 - Further motor electronics



The shaft is first produced by casting, then by pressing, rolling, or forging and finally by machining.



Impact extrusion



Graphic: Schuler

Thermal treatment



Graphic: Hirschvogel

Machining / finishing

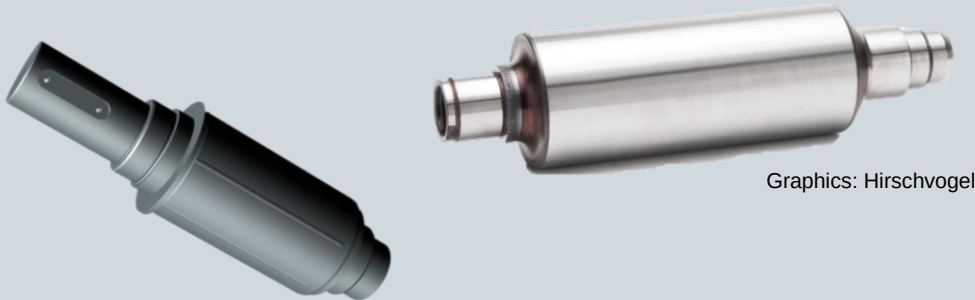


Graphic: Hirschvogel

The shaft of an electric motor transmits the force generated in the air gap as torque and can be solid or hollow.

Solid shaft

- Power transmission from air gap to output
- Supporting the rotor
- Wide selection of materials



Graphics: Hirschvogel

Advantages

- Simpler manufacturing process
- Higher strength
- Higher heat capacity

Disadvantages

- Higher weight
- Greater inertia
- Larger masses and bigger equipment required for balancing

Hollow shaft

- Special design
- Increases (thermal) design flexibility
- Strength, durability, and ductility must be ensured



Graphics: Hirschvogel

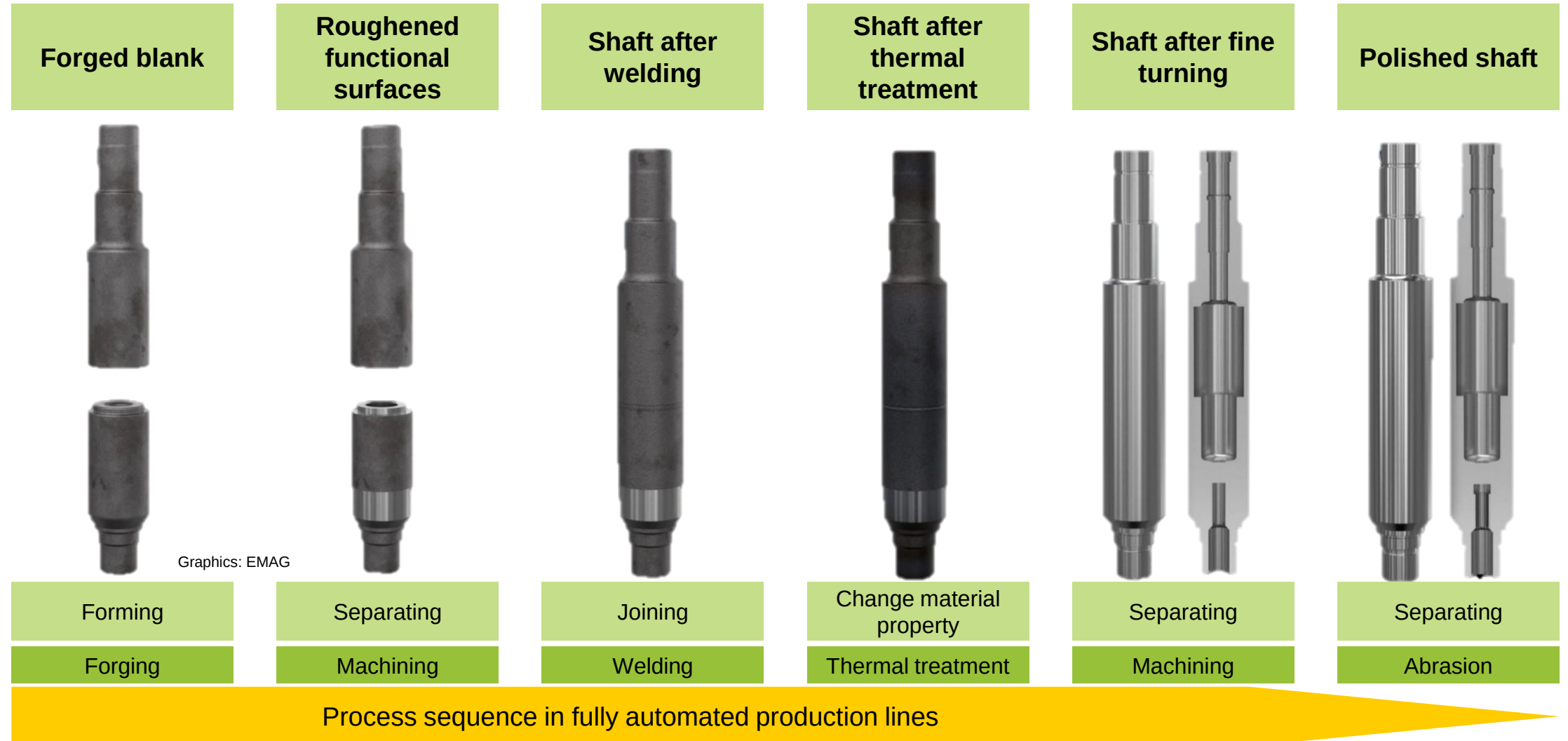
Advantages

- Lower weight and therefore simplified handling
- Possibility to implement other shafts
- (Cooling) media insertion possible

Disadvantages

- Complex, multi-stage manufacturing process
- Less choice of connection geometry

Compared to solid shafts, the production of hollow shafts is more complex and must be carried out in several parts and in several production steps.



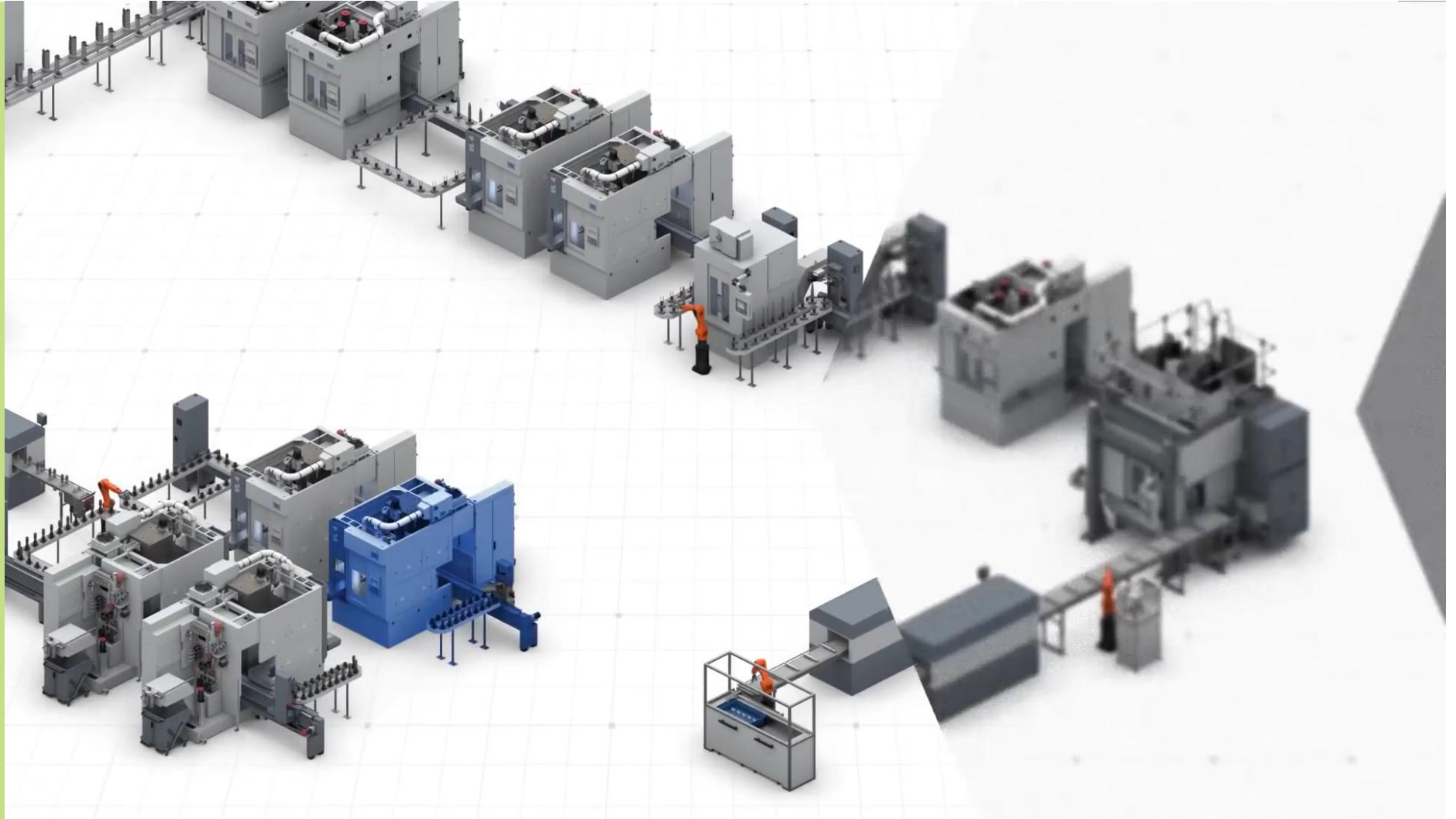
The production process of a hollow shaft can be fully automated by using multi-stage machine tools.

Process

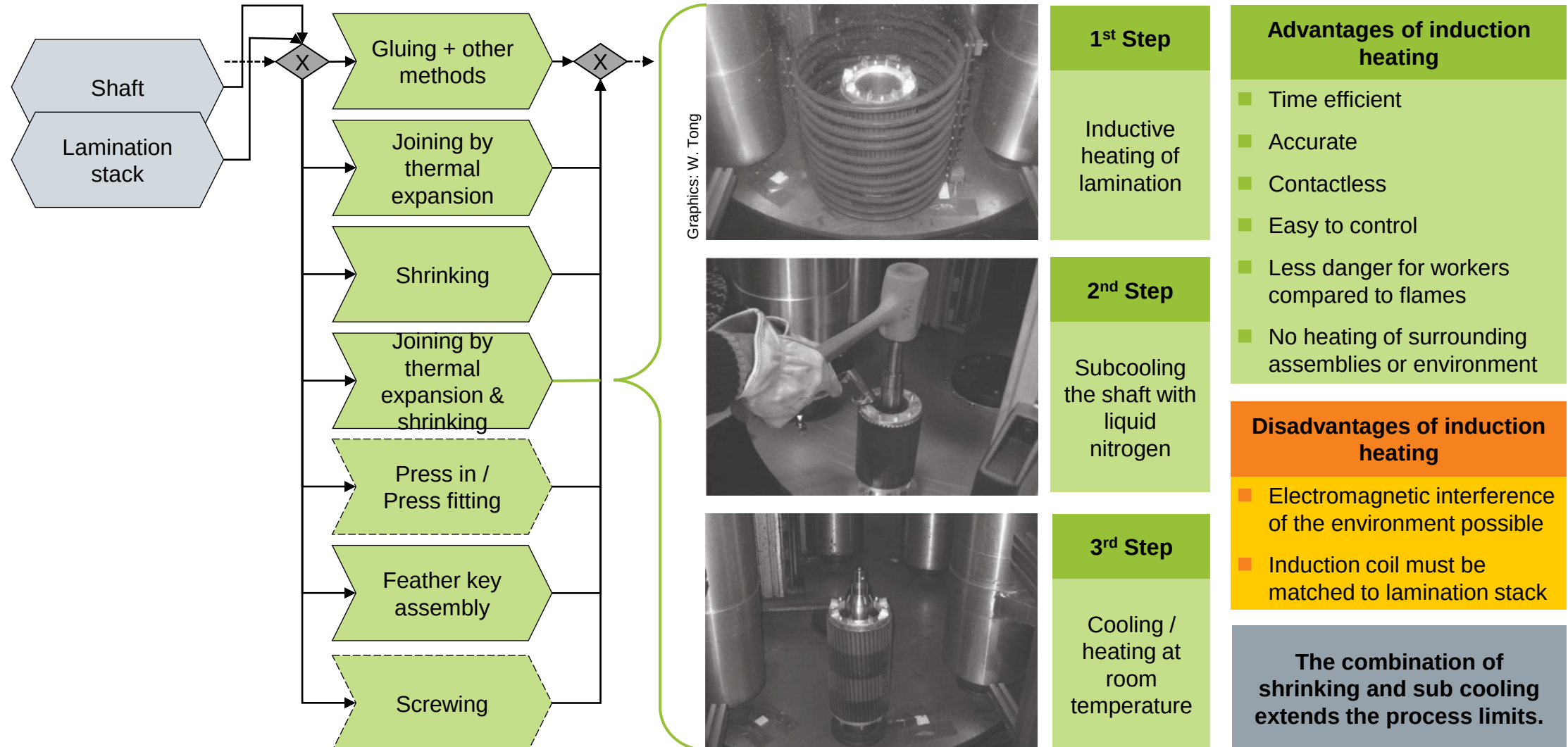
Process sequence for the construction of a two-part hollow shaft

Video: EMAG

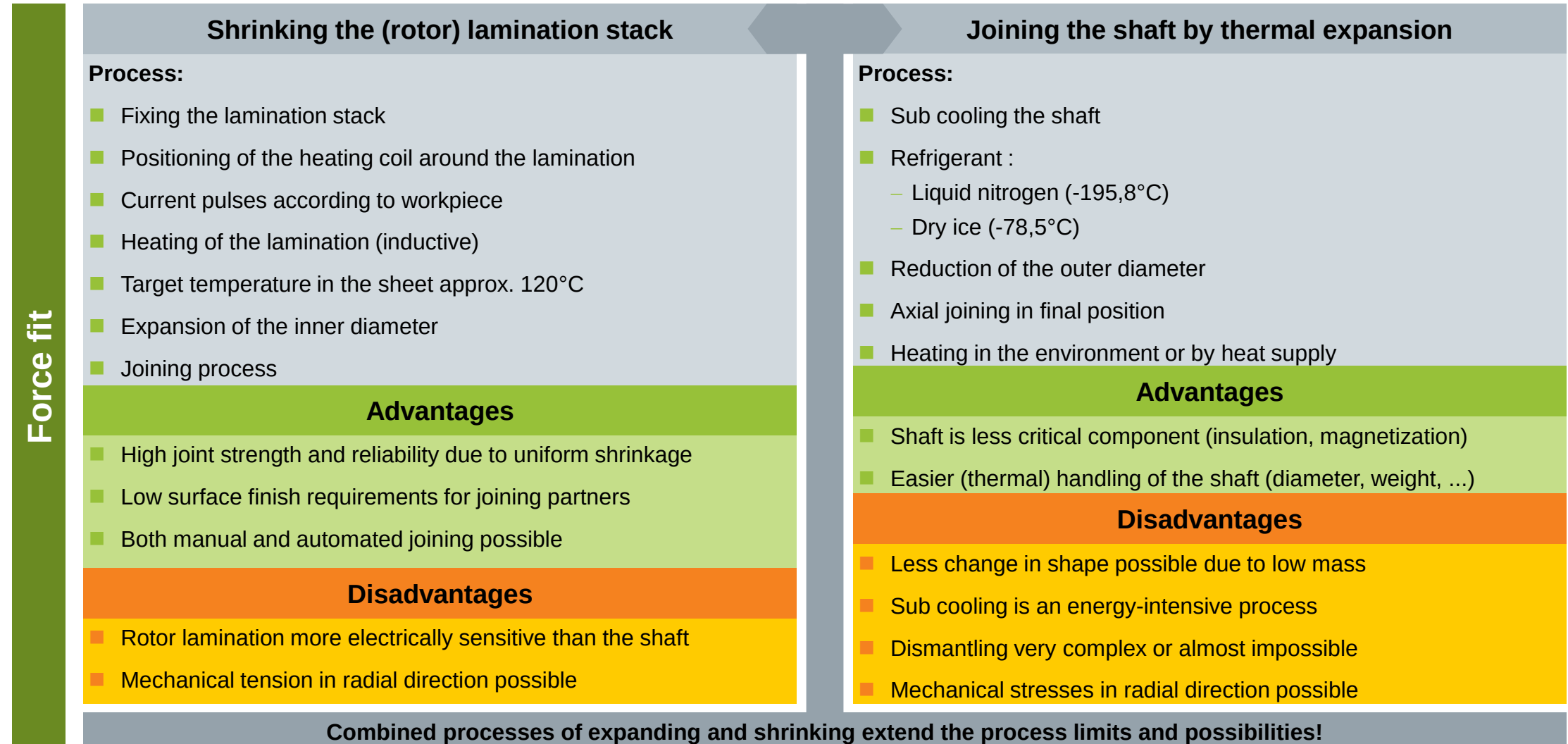
https://www.youtube.com/watch?v=HR_8_VrmmlQ



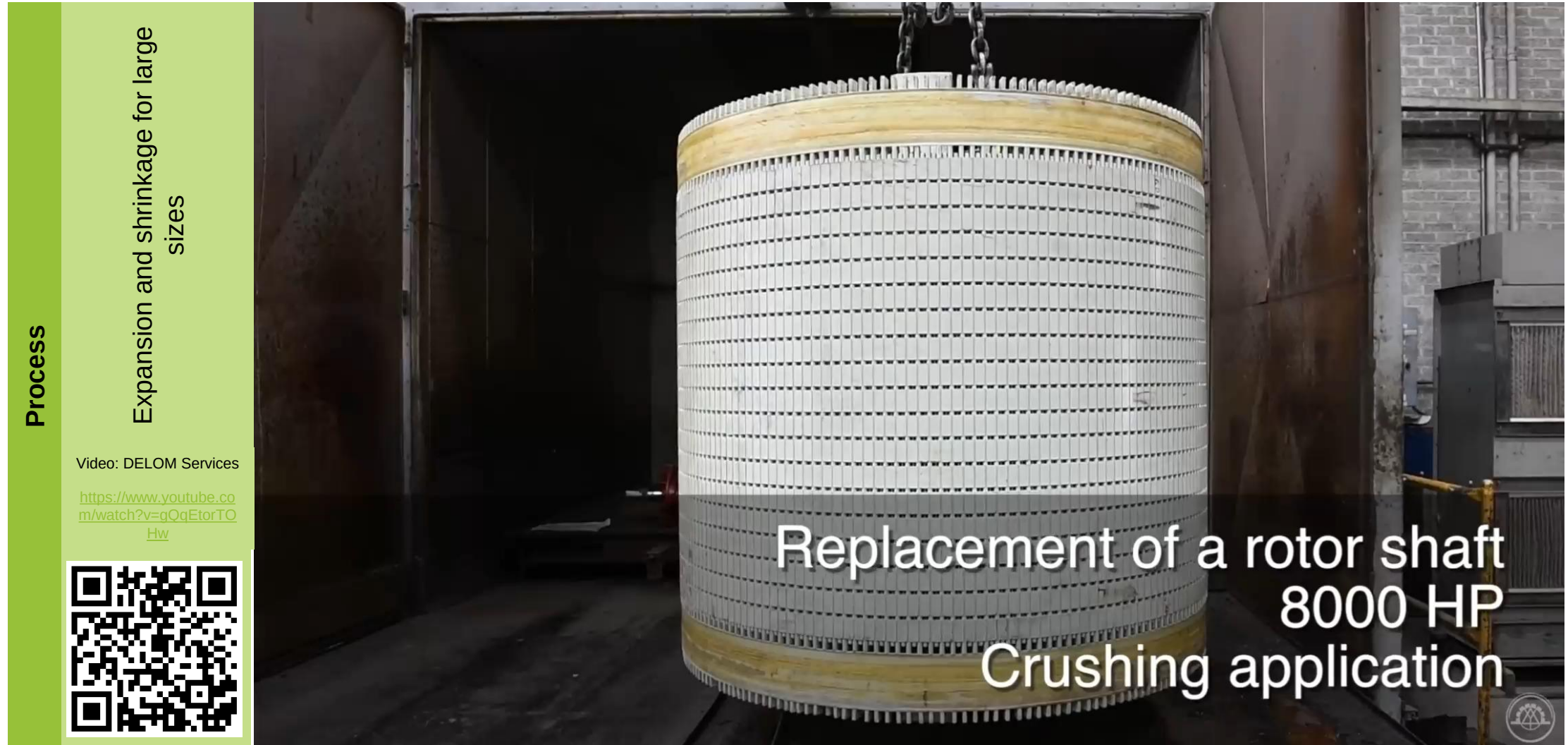
The joining of shaft and lamination by a thermal cross-press connection is the predominant method for traction drive applications.



Shrinking the laminated core, expanding the shaft, or combined methods are common ways of joining the shaft and laminated core.



The combination of expanding and shrinking increases the load-bearing capacity of the shaft-hub connection and enables higher torques to be transmitted.



Cold press-fitting of the shaft is a popular alternative process, particularly for small and simple shaft geometries.

Force fit

Press-fitting

Application:

- Small motor geometries
- Small series and prototype construction

Process:

- Axial positioning of the joining partners to each other
- Joining the shaft using axial pressure on the face (interference fit)
 - Manual press
 - Automation by using robotics
- Path-controlled process control until end position reached
- Shaft / hub contact tension retains position

Advantages

- Low costs
- Lower thermal load during press-fitting (friction)

Disadvantages

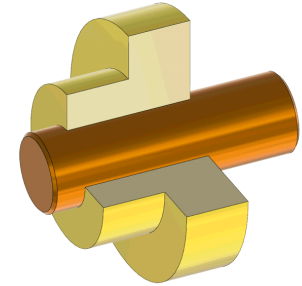
- Minor surface damage to shaft and laminated core likely
- High force required for large diameters



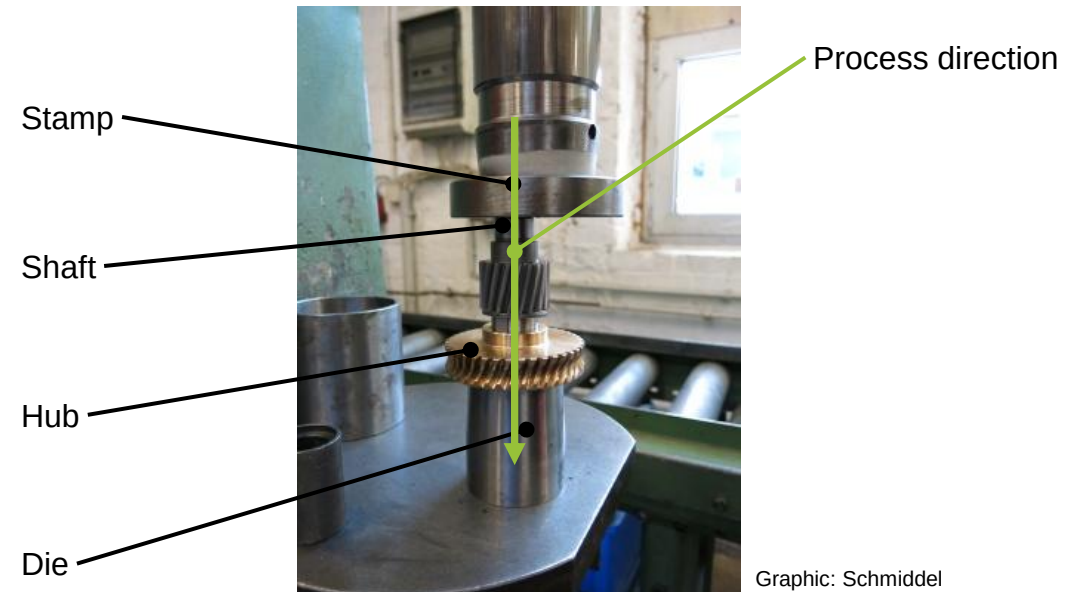
Graphic: RS Werkzeuge



Graphic: Schmidt technology

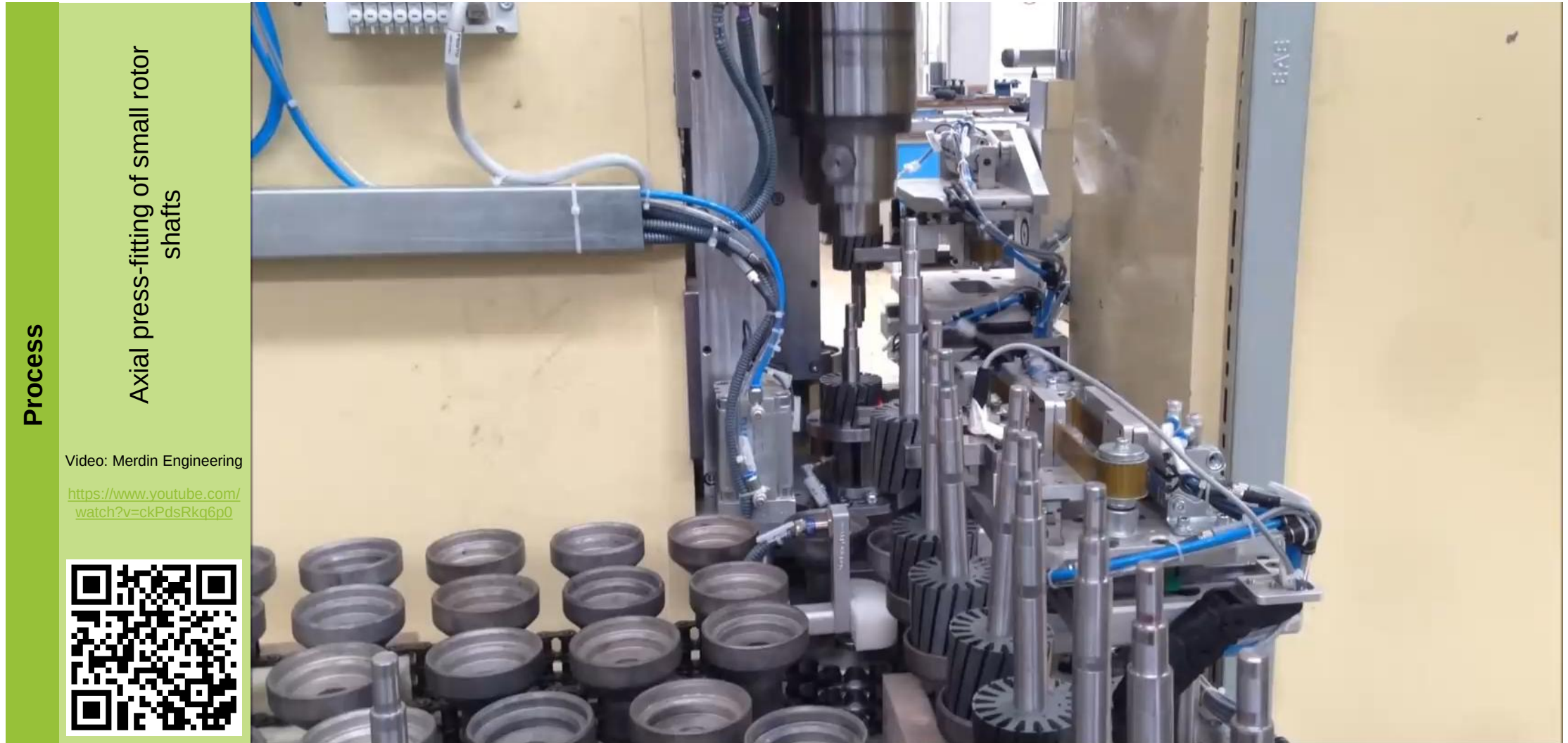


Graphic: konstruktionsatlas.de



Graphic: Schmiddel

Without energy-intensive heating or cooling of one or both joining partners, the process cycle time and thus productivity can be significantly increased.



In addition to thermally assisted processes and cold press-fitting, mechanical screwing is a common technology, primarily in the case of reluctance machines.

Form fit

Axial screwing

Application:

Reluctance machines and prototype construction

Process:

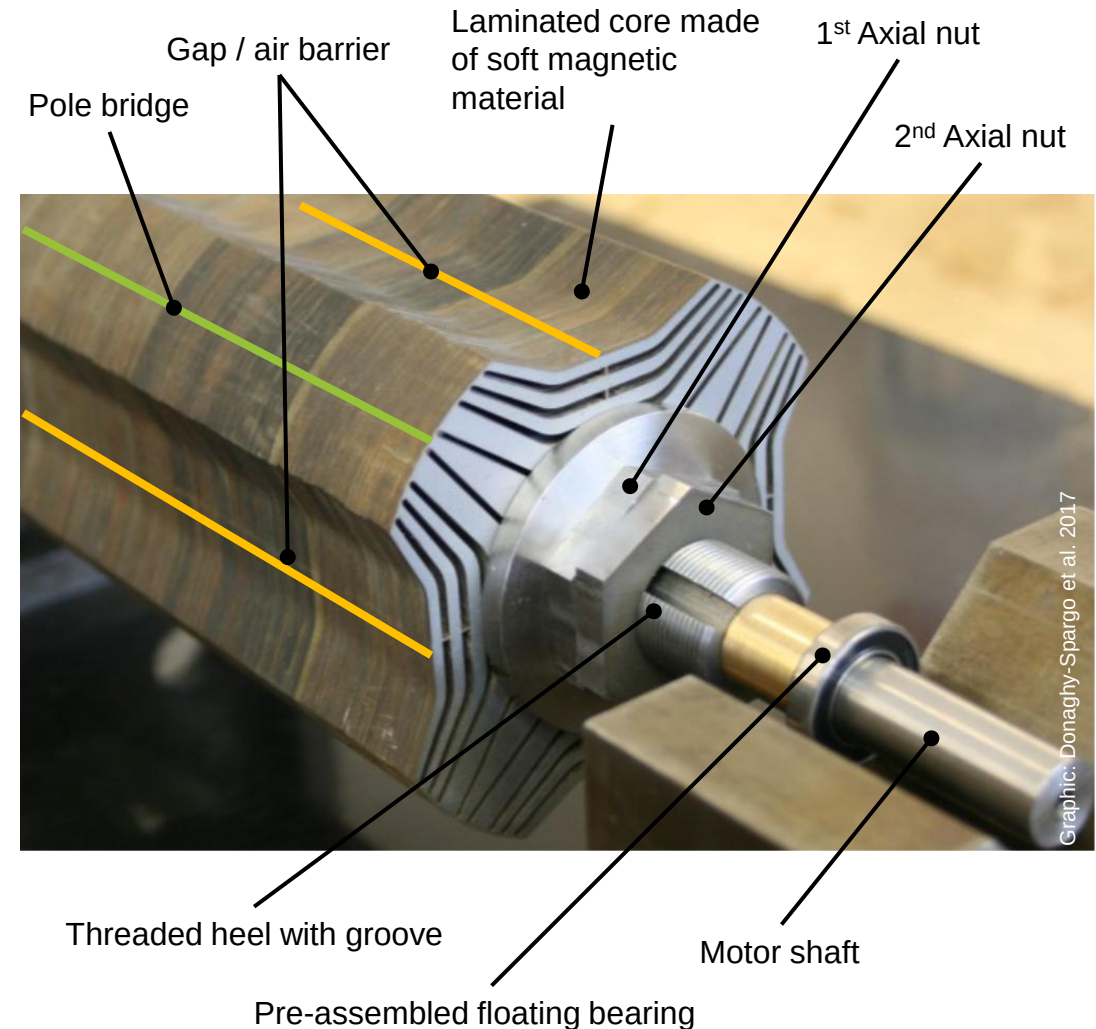
- Stacking the assembled electrical sheets to form the rotor pack
- Securing the unit
- Simple joining of the rotor shaft (no oversize fitting required)
- Torque-controlled tightening of the two axial nuts

Advantages

- Non-destructive release
- Inexpensive process without expensive machine technology (see shrink-fitting)
- No mechanical stresses in the radial direction due to remanent forces, which can lead to performance degradation (compare thermal processes or interference fit)

Disadvantages

- Possible eddy current losses from the nuts
- Damage to the laminated core possible due to incorrect assembly

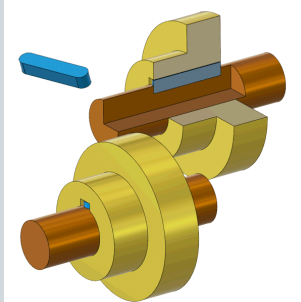


In order to transfer the force generated in the air gap to the shaft, standard machine elements can also be used depending on the expected loading.

Form fit

Feather key connection

- Keyway in shaft
- Axial joining without force
- Standardized feather key transmits force from rotor to shaft



Graphic: konstruktionsatlas.de

Advantages

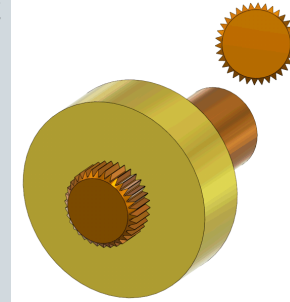
- Widely used and good standardization
- Suitable for high torques
- Non-destructive release

Disadvantages

- Risk of cracking with alternating loads
- Eccentric weight distribution
- Additional axial securing necessary

Internal / External gearing

- Multi-tooth profile at the shaft end and in the rotor hub
- Axial joining without force
- Tooth flanks act as torque transmitters



Graphic: konstruktionsatlas.de

Advantages

- Widely used and good standardization
- Suitable for solid and hollow shafts
- Non-destructive release
- Good torque transmission due to large number of transmission points

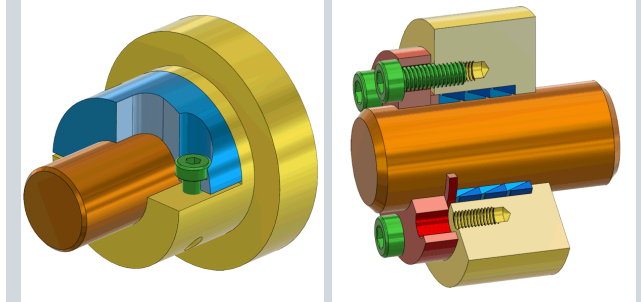
Disadvantages

- Complex manufacturing process
- Additional axial securing necessary

Force fit

Clamping / Tensioning connection

- Numerous variants available for tensioning the shaft and hub



Graphic: konstruktionsatlas.de

Advantages

- No weakening of the cross-section
- Possibility to insert other shafts
- Non-destructive release

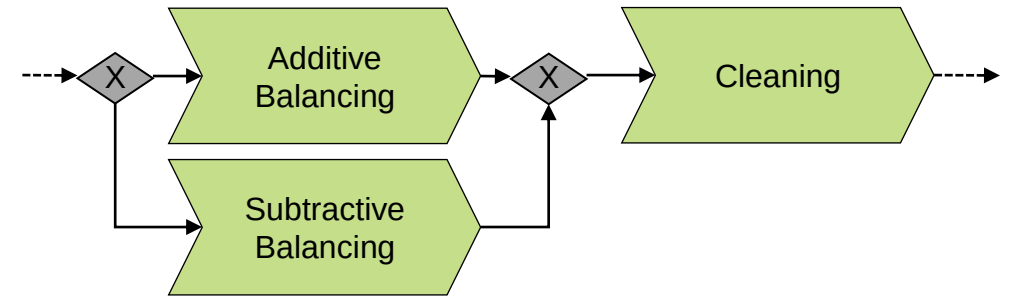
Disadvantages

- Difficult to determine an absolute position in the axial direction
- Large additional masses
- Possibly eccentric weight distribution

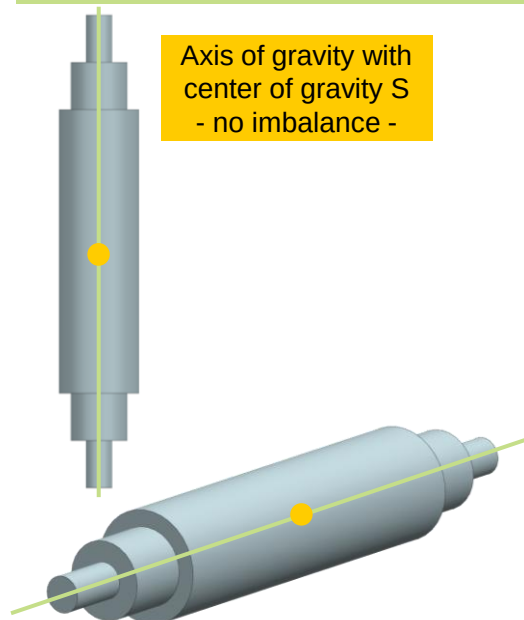
Manufacturing tolerances lead to eccentric shifts in the center of gravity, which can cause static and / or dynamic imbalance.

Imbalance

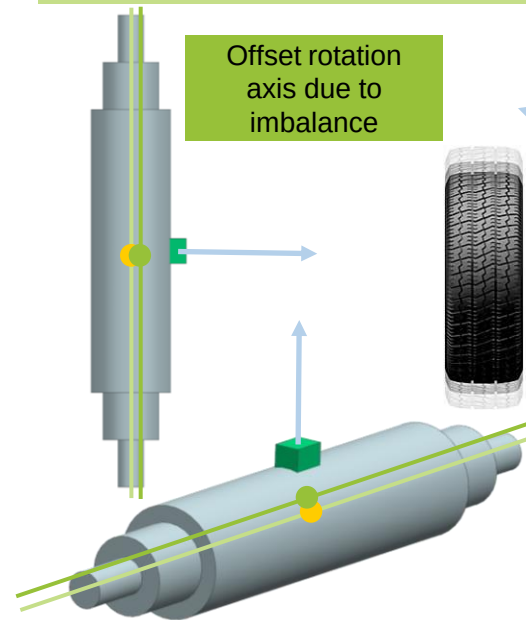
- At critical speeds, unbalance forces excite the entire vibrating system
- Vibrations, noise and increased wear are the results of unbalance
- Balancing masses to reduce these forces is called balancing



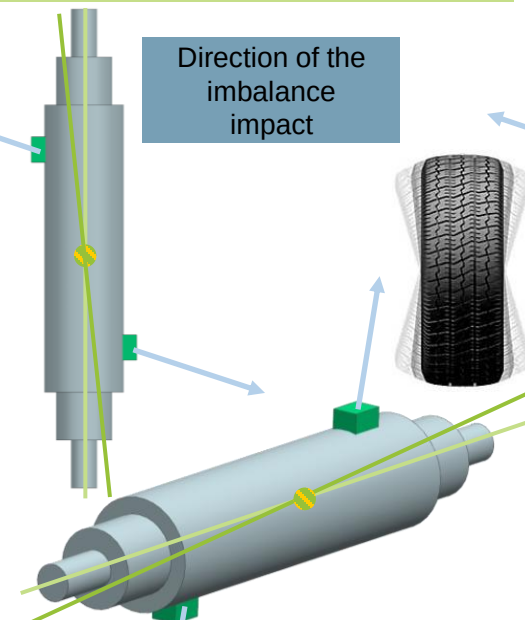
No (measurable) imbalance



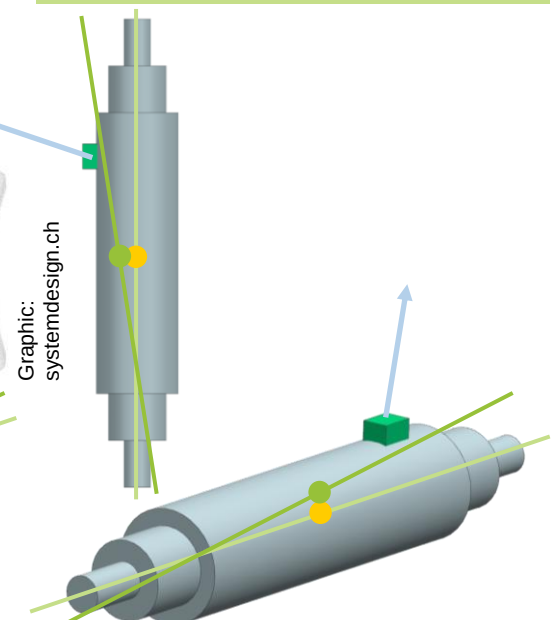
Static Imbalance No dynamic imbalance



Dynamic imbalance No static imbalance



Static imbalance AND dynamic imbalance



Balancing stations are used to determine the unbalance of one plane (static) or of two planes (dynamic) and correct it through targeted countermeasures.

Positive / Additive Balancing

Application of compensating masses, e.g. welding, soldering, gluing or screwing on weights/masses

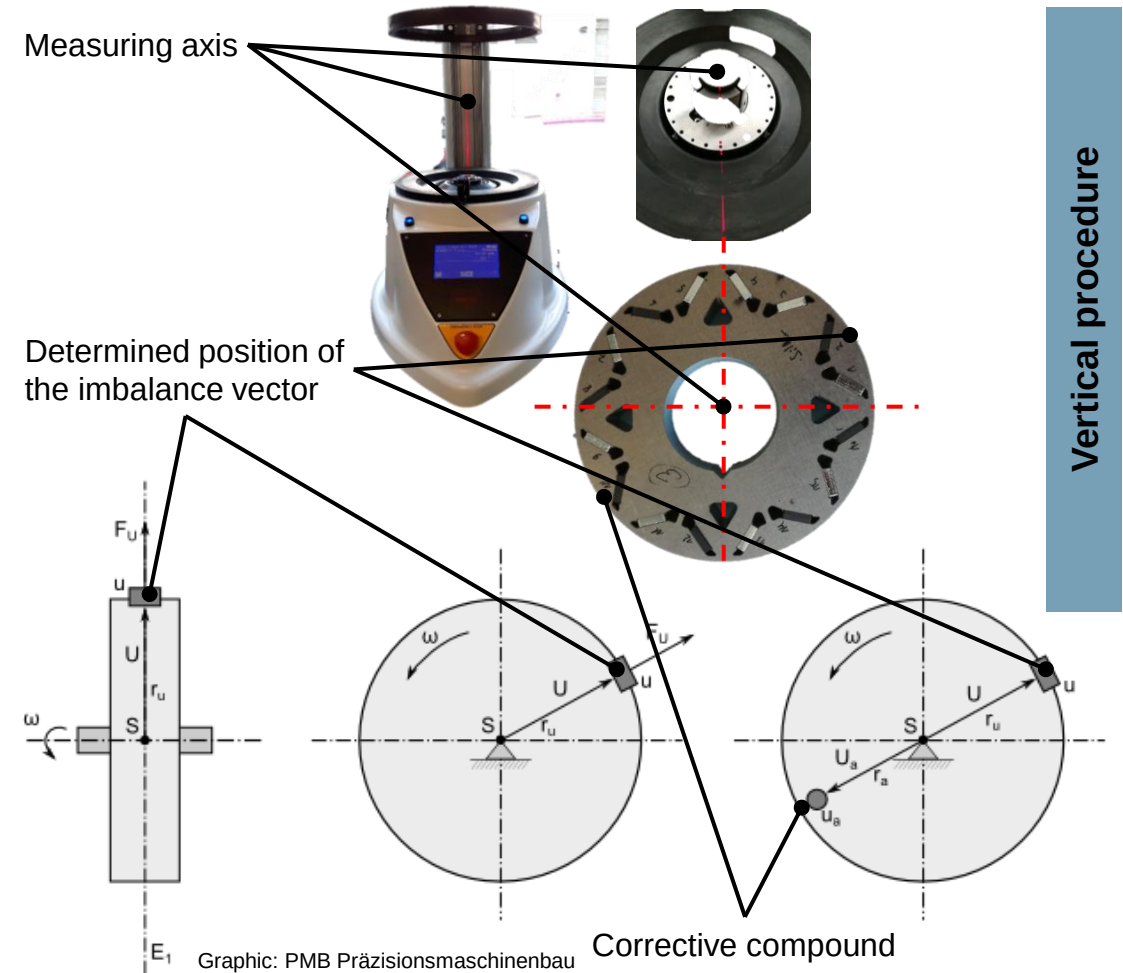
- Balancing machine sets the body to be tested (rotor) in rotation
- Mass imbalance generates centrifugal forces
- Measuring axis analytically records vector and magnitude of the forces
- Calculation of location and mass to compensate for the imbalance
- Mass compensation
- → Compare balancing process for car tires

Advantages

- Cost-effective process
- Versatile and can be used regardless of size
- Revisable in the event of process or measurement errors

Disadvantages

- Additional mass
- Larger air gap required in the machine
- Danger due to loosening of the additional mass



Balancing stations are used to determine the unbalance of one plane (static) or of two planes (dynamic) and correct it through targeted countermeasures.

Process

Additive Balancing

Video: JP Balancing

<https://www.youtube.com/watch?v=bsmPMkSNHzg>



In addition to additive processes, it is also possible to eliminate the imbalance using subtractive methods by removing material at specific points.

Negative/subtractive balancing

Removal of mass by drilling, grinding or milling

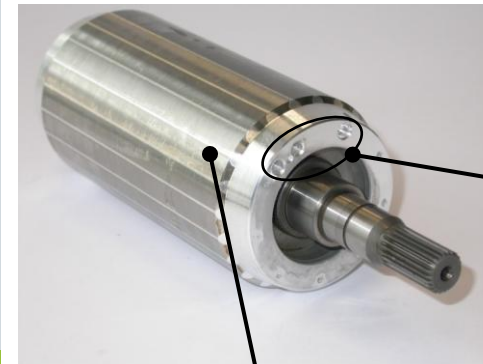
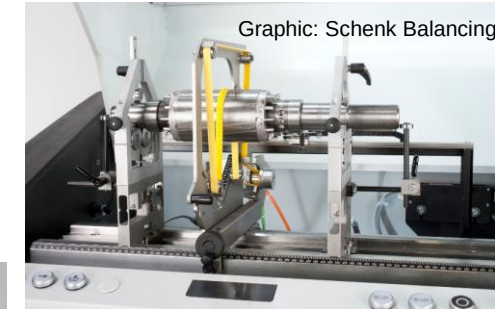
- Balancing machine sets the body to be tested (rotor) in rotation
- Mass imbalance generates centrifugal forces
- Measuring axis analytically records vector and magnitude of the forces
- Calculation of location and mass to compensate for the imbalance
- Determination of the mass to be removed
- Removal of material in place of the unbalance

Advantages

- No additional mass
- No increase in the outer diameter
- No risk of detachment
- Contactless (e.g. by laser) possible

Disadvantages

- Mistakes made by the machine or the operator cannot be undone without further ado



Subtractive balancing by means of drilling



Manual removal of the imbalance

Rotor lamination

Horizontal procedure

In addition to additive processes, it is also possible to eliminate the imbalance using subtractive methods by removing material at specific points.

Process

Subtractive balancing by drilling

Video: ICARinc

<https://www.youtube.com/watch?v=ZBAAt1yxuec>



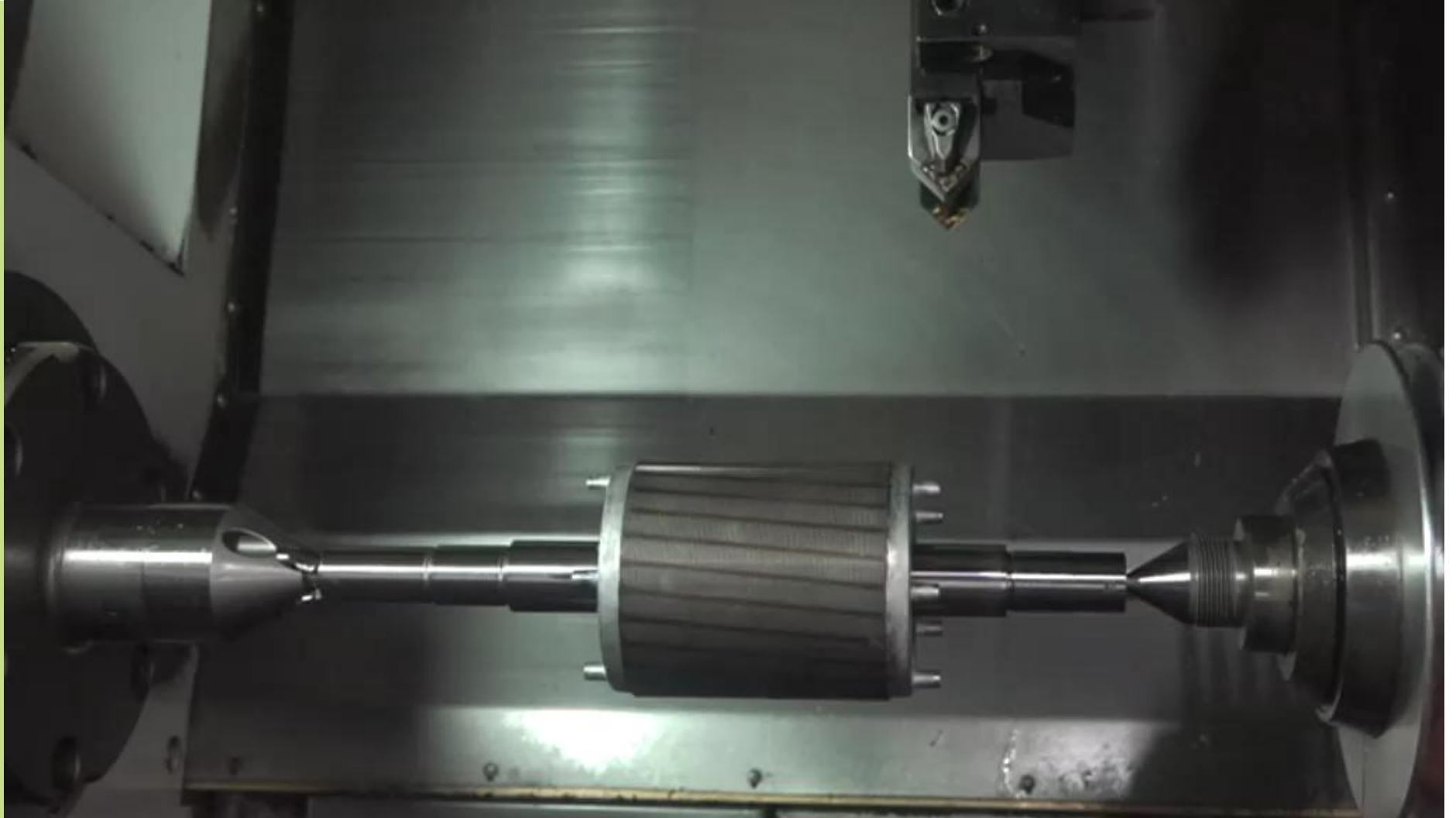
Fully automated execution of all production processes is essential, especially for large quantities.

Process

Machining and balancing of the rotor shaft

Video: Siemens

<https://www.youtube.com/watch?v=p8MeR8IT3GA>

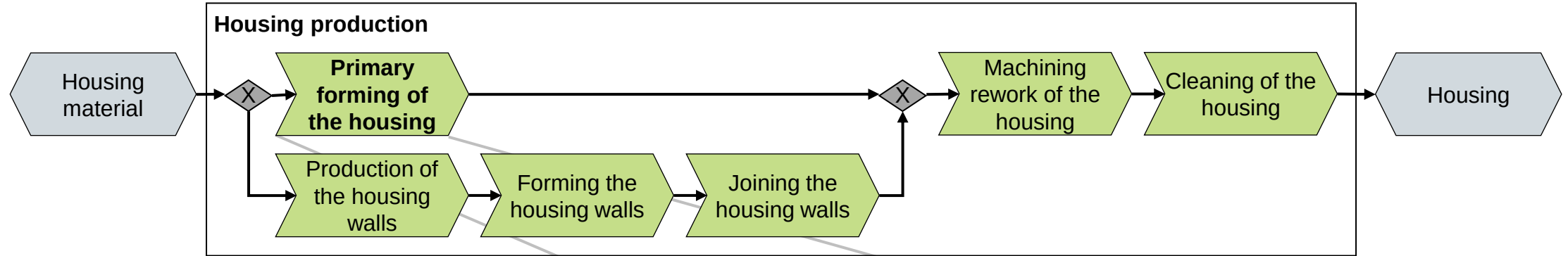


This unit covers the assembly processes used in the manufacture of electrical drives for components that do not directly affect electromagnetic energy conversion.

- Motivation and positioning of assembly within the process chain
- Pre-assembly of the rotor shaft
 - Production and assembly of shaft and rotor
 - Balancing the rotor shaft
- **Pre-assembly of stator and housing**
 - Production and joining process of stator and housing
 - Mounting the bearing shield
- Final assembly of the modules
 - Joining rotor shaft and stator
 - Further motor electronics



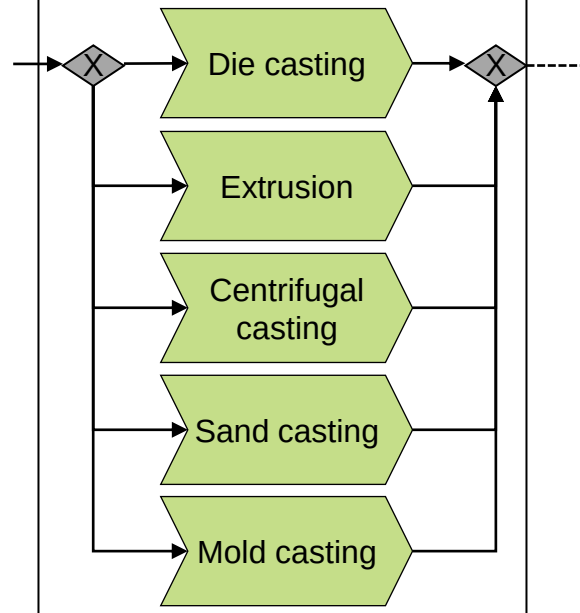
The housing of an electric motor fulfills various tasks and is usually produced using conventional manufacturing processes.



Tasks and requirements for the housing

- Torque support
- Adjusting and maintaining the coaxiality of the rotor and stator
- Securing and centering the bearing end shields
- Mounting for the power and signal connections
- Heat dissipation
- Protecting the active components from environmental influences
- Transport protection and positioning of the motor at the installation site

Primary forming of the housing



Graphic: Trompetter GUSS



Depending on the size, design and intended use of the electric machine, a wide range of housing topographies are available.

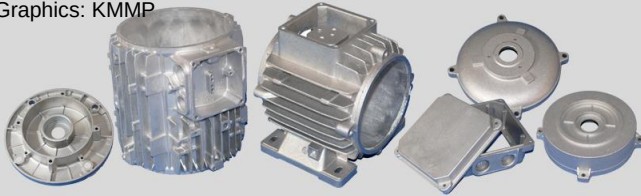
Casted housing

DIN 8580 – Primary forming

Process:

- Die casting
- Sand casting
- Cf. technical literature or lecture PTI/PTII

Graphics: KMMP



Materials:

- Cast iron
- Aluminum

Advantage

- Suitable for large quantities
- Cooling channels / ribs can be integrated
- Size flexibility

Disadvantage

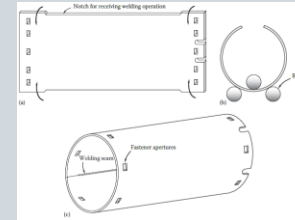
- High costs for machinery
- Machining rework required
- Not suitable for small series

Wrapped sheet metal housing

DIN 8580 – Forming

Process:

- Pre-punching of the sheet geometry
- 3-roll round bending forming
- Assembly of all motor components
- Welding of the two joint edges



Graphics: Transmotec, Wei Tong



Advantage

- Very cost-effective production of semi-finished products
- Low forming force required

Disadvantage

- Low stability due to thin-walled construction
- Low heat dissipation
- Additional thermal load due to welding process

Milled housing

DIN 8580 – Separating

- Usual procedure: machining
- Cf. technical literature or lecture PTI/PTII



Graphic: DMG Mori

Advantage

- Suitability for rapid prototyping
- High flexibility with materials and machine tool (NC, CNC, manual)
- Precise production possible

Disadvantage

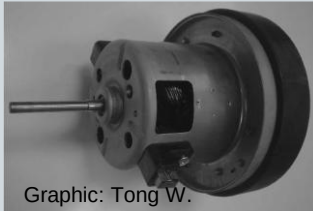
- Not suitable for large-scale production
- Not suitable for large sizes

Depending on the size, design and intended use of the electric machine, a wide range of housing topographies are available.

Punched/drawn housing

DIN 8580 – Forming

- Common procedure: Forging
- Cf. technical literature or lecture PTI/PTII
- Function integration possible: :
 - Bearing shield
 - Bearing seat
 - Connection



Advantages

- Suitable for large quantities - cycle time eff.
- Interesting for less demanding applications due to its low cost.

Disadvantages

- High forming forces required
- Expensive machinery required
- Only suitable for small sizes

Extruded housing

DIN 8580 – Forming

Process:

- Inserting the material
- Heating the material
- Increasing the forming pressure
- Moving the punch
- Material flows through die



Advantages

- Complex cross-sections possible
- Cooling function integration
- Low weight
- Strength and stability
- Manufacturing precision

Disadvantage

- Only suitable for aluminum in EDP

Welded housing

DIN 8580 – Joining

Process:

- Cutting / reshaping the sheet metal parts
- Welding the individual parts
- Machining reworking

Usually found in low-cost countries due to the high percentage of manual labor.



Advantages

- Lucrative for low quantities
- Suitable for very large sizes
- Complex geometries possible on the inside

Disadvantages

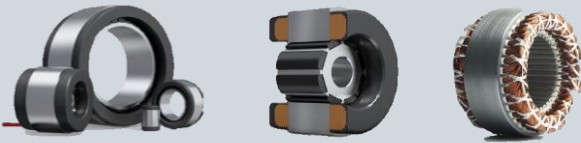
- Long production times due to many individual steps
- Automation difficult to implement (variants)

For specialized or experimental designs, further technologies are available for the manufacture of stator housings, some of them are still under development.

Housingless variants

No production of a separate housing

Lightweight construction promotes the use of housingless variants and thus enables use in a wider range of products



Graphics: Wittenstein

Advantages

- Reduced weight and volume
- Reduction of the required installation space
- Greatest possible freedom of integration
- Reliability and efficiency
- Savings in assembly steps/costs

Disadvantage

- More complex sealing

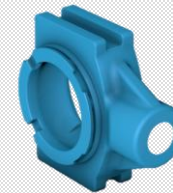
Plastic / composite housing

DIN 8580 – primary forming/forming

New manufacturing technologies and applications for electric drives are forcing the development of alternative manufacturing processes.



Graphic: Fraunhofer



Graphic: SKF

Advantages

- Weight reduction
- Low vibration propagation
- No corrosion possible

Disadvantages

- Costs
- Manufacturing technologies not yet sufficiently developed in the context of EDP

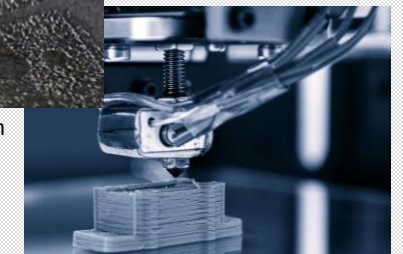
3D-Printing / Additive procedures

DIN 8580 – primary forming



Graphic: industrieller.com

Graphic: mittelstand-heute.com



Possible technologies

Plastics

- 3D-printing
- Selective laser sintering
- Stereolithographic
- Thermal transfer sintering

Metal

- 3D-printing
- Selective laser sintering
- Laser beam melting
- Electron beam melting

The motor housing is usually shrunk onto the stator to provide good axial alignment and heat transfer.

Shrinking

Process:

- Secure the housing
- Stator laminated core positioned axially to the housing
- Current pulses heat up the housing
- Expansion of the inside diameter
- Joining process in axial direction with reduced force required

Advantages

- High joint strength and reliability due to uniform shrinkage
- Low surface requirements for joining partners
- Both manual and automated joining possible

Disadvantage

- Energy intensive procedure

Press-fit

- Only used when it is not possible or necessary to heat the housing
- Especially for:
 - Small motors
 - Motors with low power density
 - Very short cycle times
 - Mass-produced goods or
 - Budget goods

Advantages

- Cost-effective processing
- Short cycle time
- Less energy intensive

Disadvantages

- Possible damage to the housing / stator must be accepted
- Poor heat dissipation due to grooves and scratches
- Comparatively weak hold in the housing

Gluing

- Large selection of adhesive variants
- See technical literature or lecture PTI/PTII
- Mostly used in addition



Advantages

- Compensation of larger tolerances
- No frictional or contact corrosion
- Impact resistant, damping, sealing
- Homogeneous stress distribution

Disadvantages

- Sudden failure possible
- Influence of operating temperature

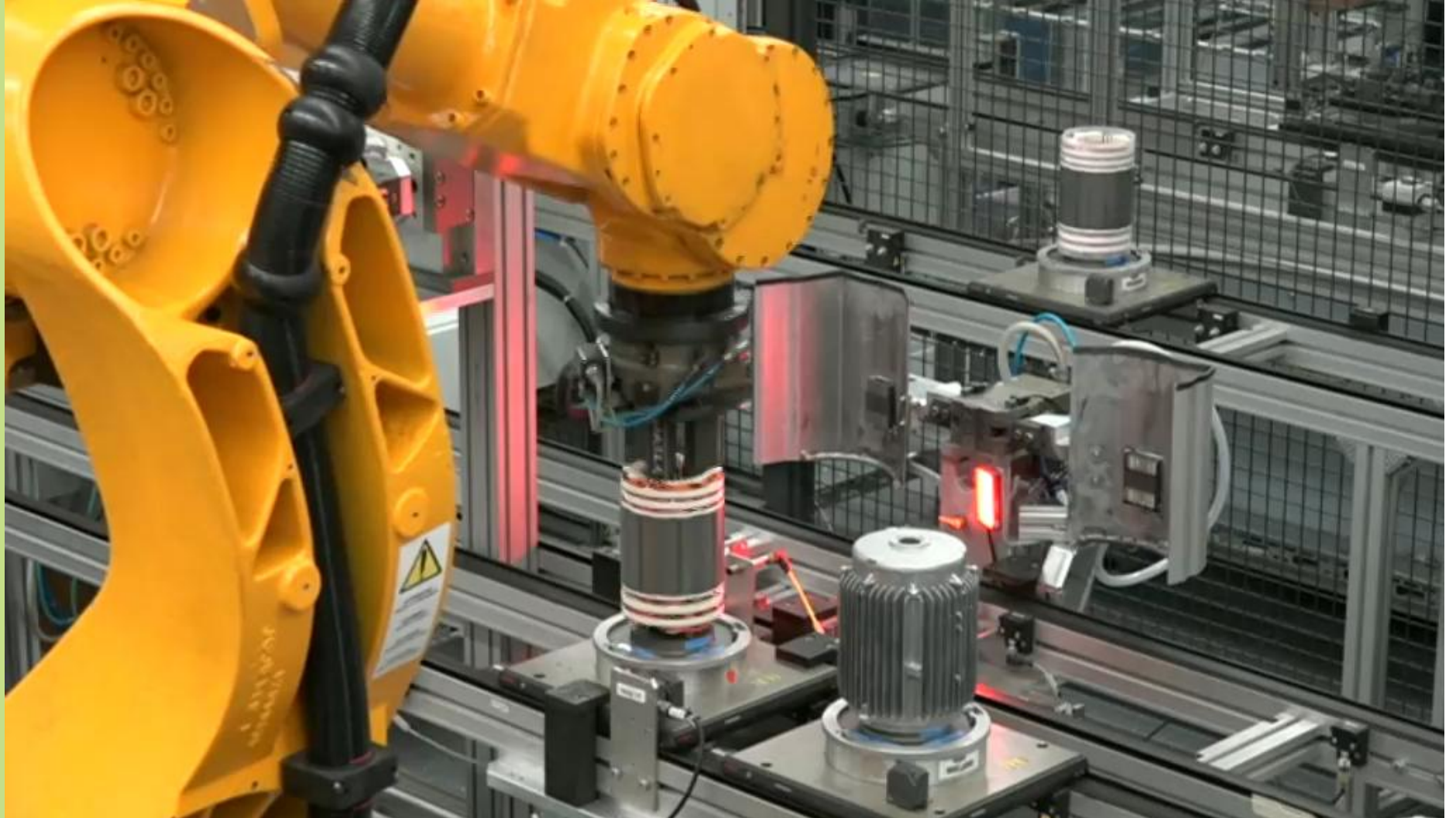
The motor housing is usually shrunk onto the stator to provide good axial alignment and heat transfer.

Process

Shrinking the motor housing

Video: Siemens

<https://www.youtube.com/watch?v=p8MeR8IT3GA>



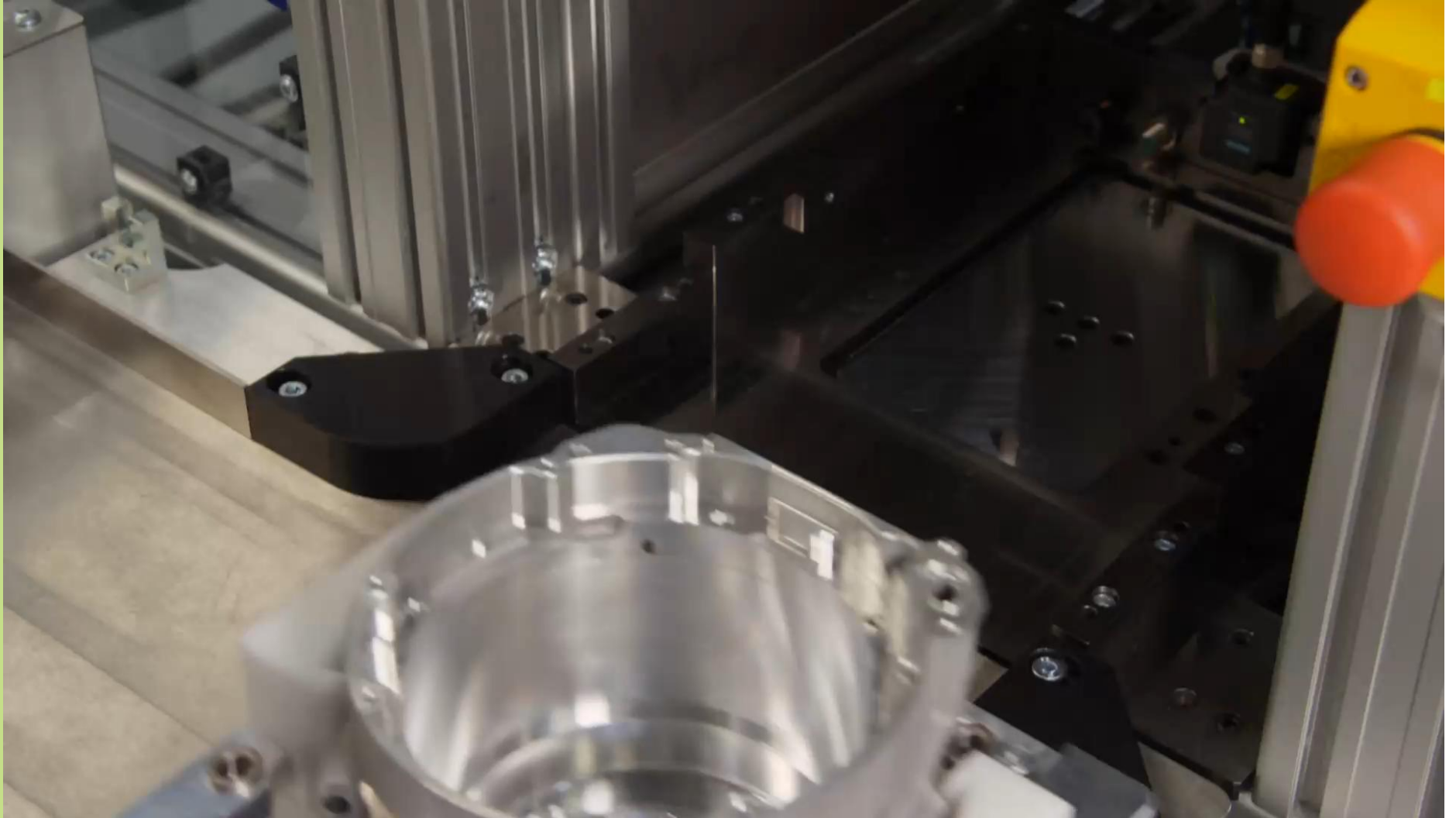
The motor housing is usually shrunk onto the stator to provide good axial alignment and heat transfer.

Process

Shrinking the motor housing

Video: Continental AG

<https://www.youtube.com/watch?v=ft04LbFXRZE>



To ensure smooth, low-vibration operation, the rotor shaft must be guided by bearings on both sides with the utmost precision.

Bearing

- Supporting and guiding parts that move relative to each other
- Absorption of forces in the radial direction
- If necessary, absorption of forces in the axial direction
- Guarantee low-friction transfer of the torque
- Standardized design elements

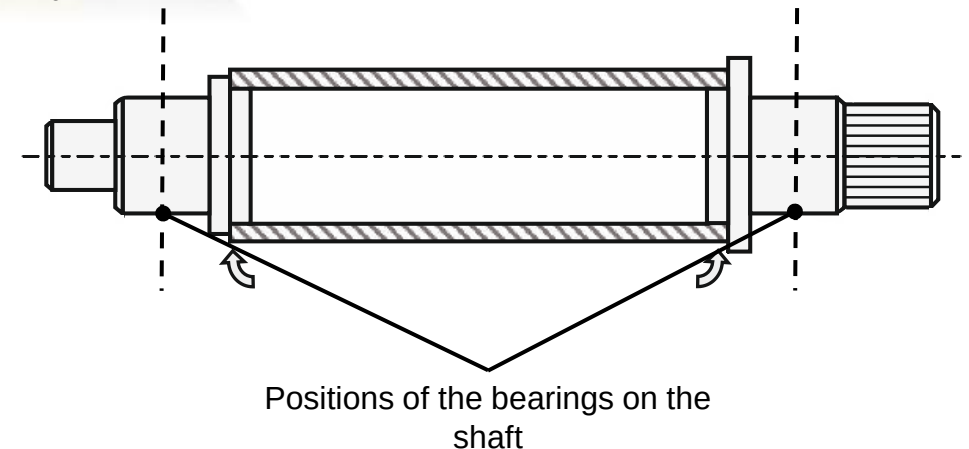
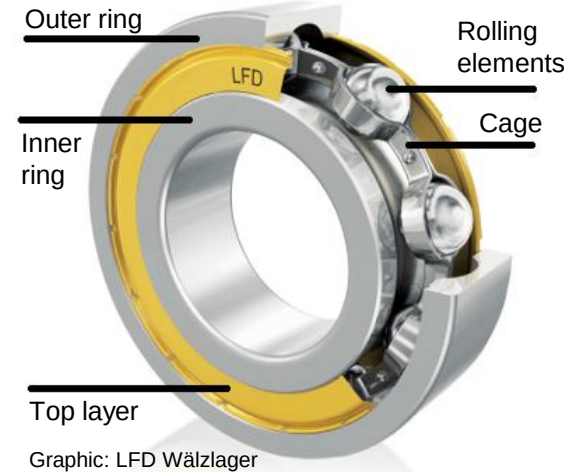
Types of bearings

- Radial ball bearings
- Radial roller bearings
- Axial ball bearings
- Axial roller bearings
- Spherical plain bearing



Additional requirements

- Sufficiently rigid bearing seats
- Axial fixation
- Lubrication if necessary
- Maintenance and inspection



Depending on the bearing type, the bearing size and the expected load, various methods are available for connecting the bearing to the seat or shaft.

Bearing assembly

- Fit must be determined in advance
- Various process designs possible
- Can be implemented manually or automatically
- By applying pressure to the outer ring of the bearing
- Bearing and seat must always be coaxial
- Use of lubricant facilitates process control



Joining processes

- Mechanical procedures:
 - Press in/press on with continuous force
 - Drifting in/up by impact/impact-like application of force
- Thermal procedures:
 - Heating with oil or hot air (~100°C, suitable for floating bearing)
 - Induction heating (suitable for floating bearing)
 - Sub cooling (suitable for fixed bearings)
- Hydraulic or hydraulically assisted processes

Graphic: Schaeffler AG

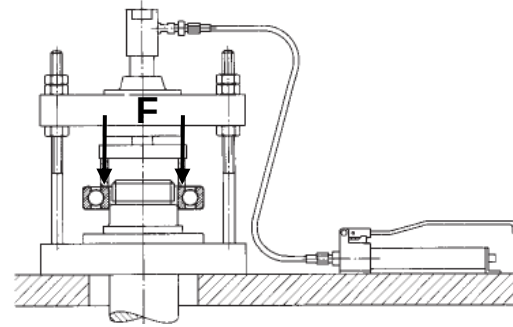
Quantity

Effort

Size

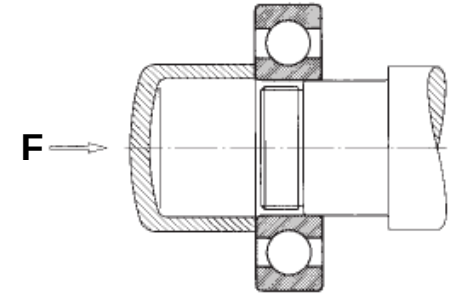
Typically, the diameter of the hole determines the joining method to be used.

Mechanical pressing



$d < 80\text{mm}$

Mechanical driving



$d < 50\text{mm}$

Inductive heating



$20\text{mm} < d < 100\text{mm}$

Direct inductive joining



$d > 100\text{mm}$

While the floating bearing is already mounted on the rotor shaft during pre-assembly, the fixed bearing is mounted in the first bearing shield.

Tasks of the bearing shields

- Closing the open motor ends
- Supporting the bearings
- Ensure cold air supply
- Protect against external influences
- Noise reduction



The properties, advantages and disadvantages of the processes are comparable to those of the manufacturing processes for housings.

Forged bearing shields

Advantages

- High load-bearing capacity due to favorable metal structure
- Heat treatment possible

Disadvantages

- Only simple geometries possible
- Time-consuming and therefore expensive

Cast bearing shields

Advantages

- Suitable for large quantities
- Cooling channels / ribs can be integrated
- Size flexibility

Disadvantages

- High costs for machines
- Machining rework required

Punched bearing shields

Advantages

- Suitable for large quantities - cycle time
- Interesting for less demanding applications due to its low cost

Disadvantages

- Only suitable for small sizes
- No high-end products possible

Machined bearing shields

Advantages

- Suitability for rapid prototyping
- Common machine tools
- Precise production possible

Disadvantages

- Not suitable for large series production
- Not suitable for large sizes

Common methods for connecting the bearing shield to the housing are screwing, gluing and pressing.

Connecting the bearing shields to the housing

- Secure the components
- Positioning of the stator lamination stack axially aligned to the bearing seat and thus to the motor shaft
- Temperature dissipation
- Prevention of external influences
- Ventilation

Press-fit using locking elements

Advantages

- Non-destructive release
- No additional thickening of the housing necessary
- Very low additional design effort
- Very low additional mass

Disadvantage

No sealing effect

Screwing

Advantages

- High joint strength and reliability due to uniform shrinkage
- Low surface requirements for joining partners
- Both manual and automated joining possible
- Non-destructive release
- Variable tightening torque

Disadvantages

- No sealing effect
- Housing must have sufficient material thickness by design
- Additional weight due to thickening

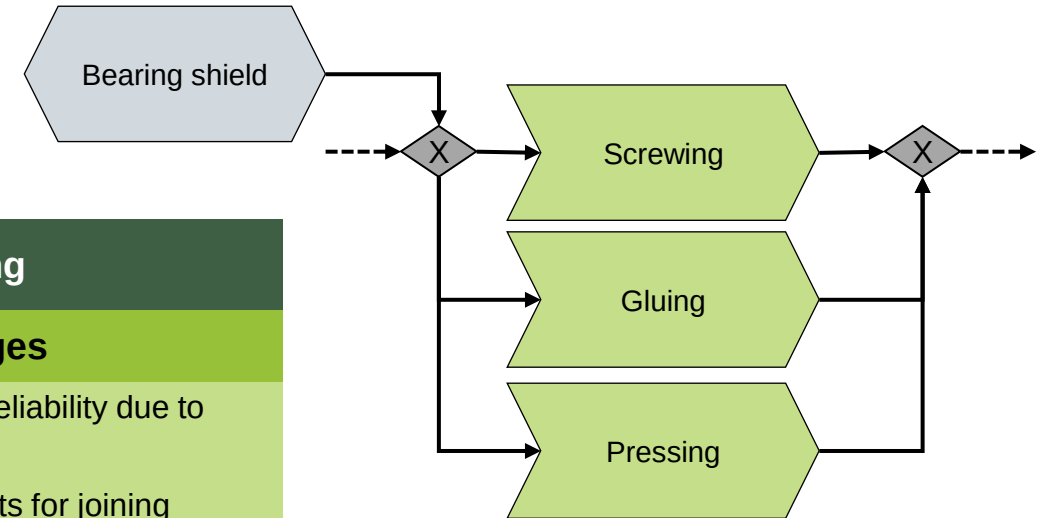
Gluing

Advantages

- Compensation for manufacturing tolerances
- No frictional or contact corrosion
- Impact resistant, damping, sealing

Disadvantages

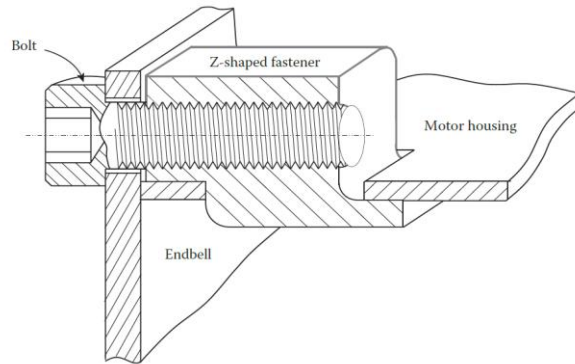
- Sudden failure possible
- Influence of operating temperature



In most applications in electromechanical engineering, locking elements are used to connect the housing and the two end shields.

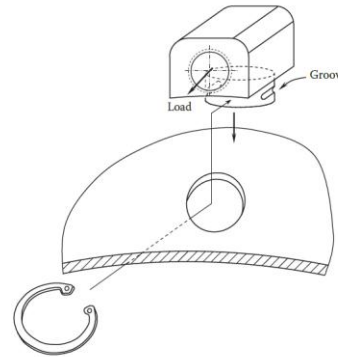
Form fit

Z-connector



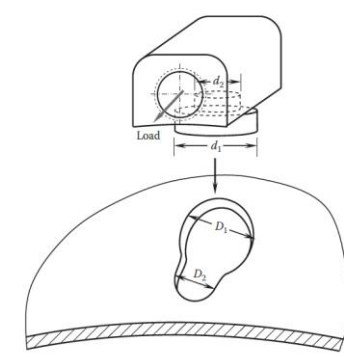
Push through & hook in

Cylindrical connector



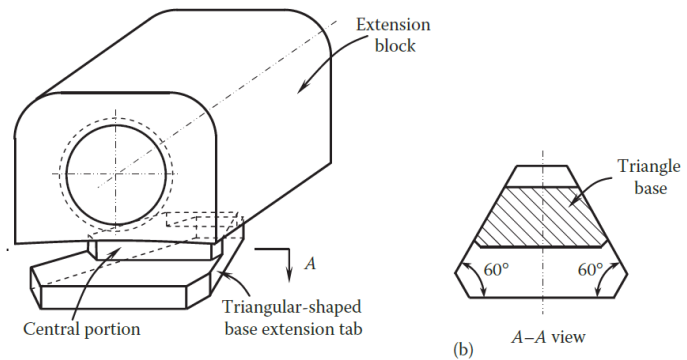
Plug in & secure

Cylindrical connector



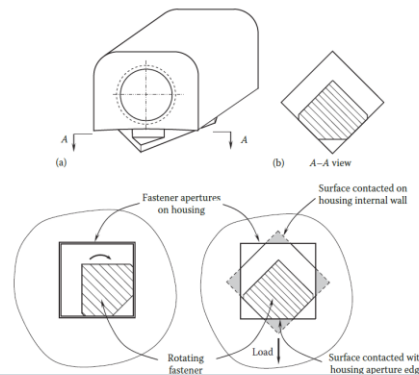
Plug in & secure

Triangular connector



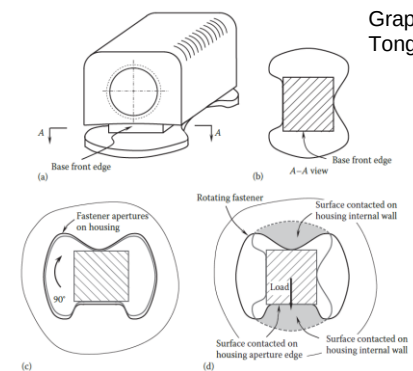
Plug in & rotate

Square connector



Plug in & rotate

Butterfly connector



Plug in & rotate

Graphics: Wei Tong

This unit covers the assembly processes used in the manufacture of electrical drives for components that do not directly affect electromagnetic energy conversion.

- Motivation and positioning of assembly within the process chain
- Pre-assembly of the rotor shaft
 - Production and assembly of shaft and rotor
 - Balancing the rotor shaft
- Pre-assembly of stator and housing
 - Production and joining process of stator and housing
 - Mounting the bearing shield
- **Final assembly of the modules**
 - Joining rotor shaft and stator
 - Further motor electronics



To seal the motor, a sealing compound can be applied to the bearing shield in addition to the screw connection used during final assembly.

Process

Bearing assembly and bearing shield assembly

Video: ABB Ltd.

<https://www.youtube.com/watch?v=CBFE-Bt7RjY>



Mainly manual assembly

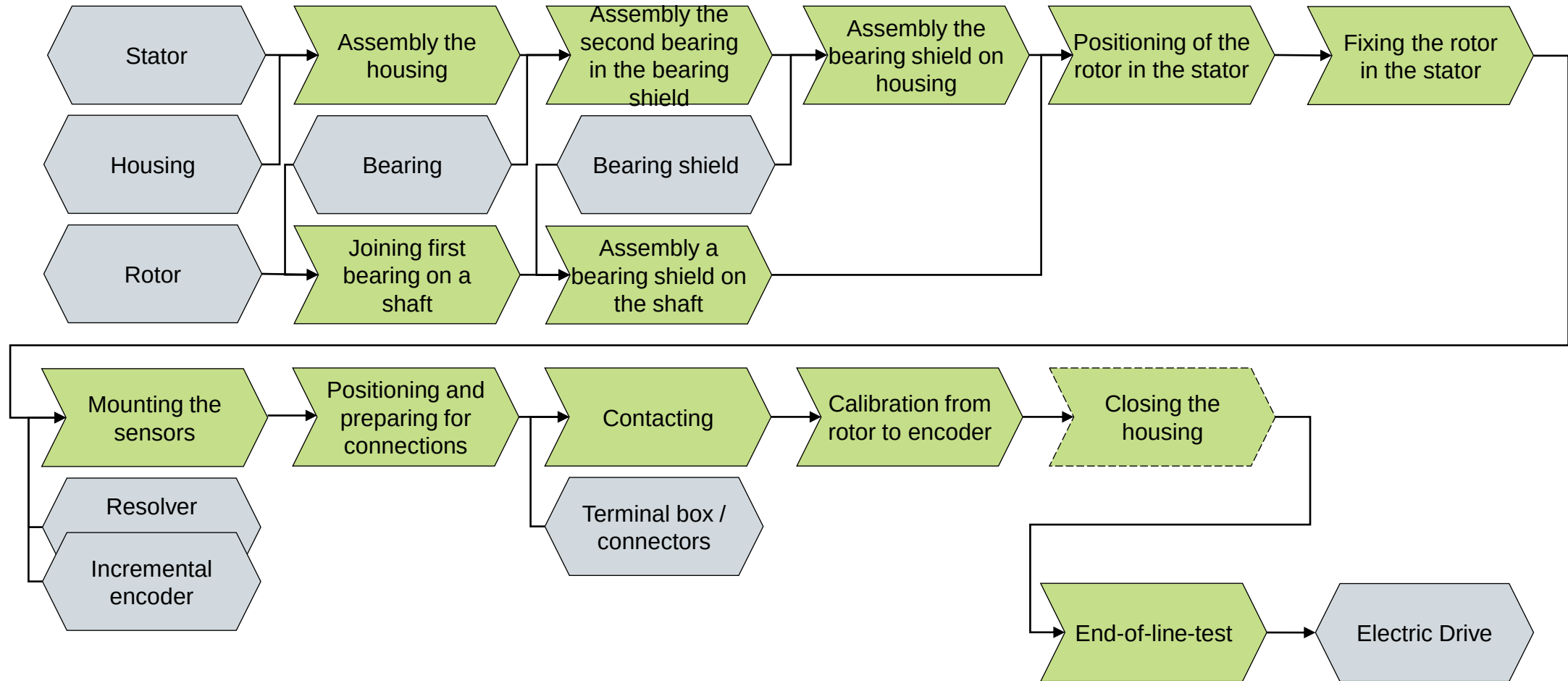
Additional work steps demonstrated:

- Joining the rotor and stator
- Packing the bearings to protect them from particles
- Screwing on the end shield

Not demonstrated works steps:

- Protective coating of rotor and stator
- Inductive heating of the bearings

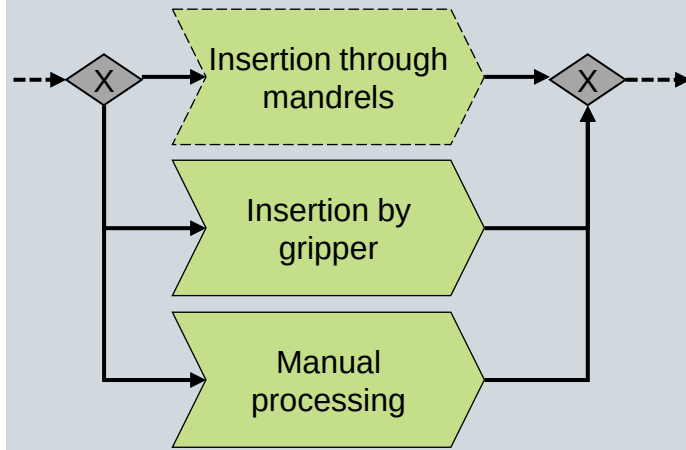
The final assembly ("marriage") of stator and rotor is followed by various assembly steps, depending on the design and functional scope of the motor.



Bonding the rotor to the stator is particularly critical with permanently excited rotors because the rotor magnets are pulled toward the stator laminations.

"Marriage" = joining of rotor and stator unit

- Coaxial alignment must be maintained
- Under no circumstances may the rotor be pulled towards the stator in the air gap and thereby damaged (PM)

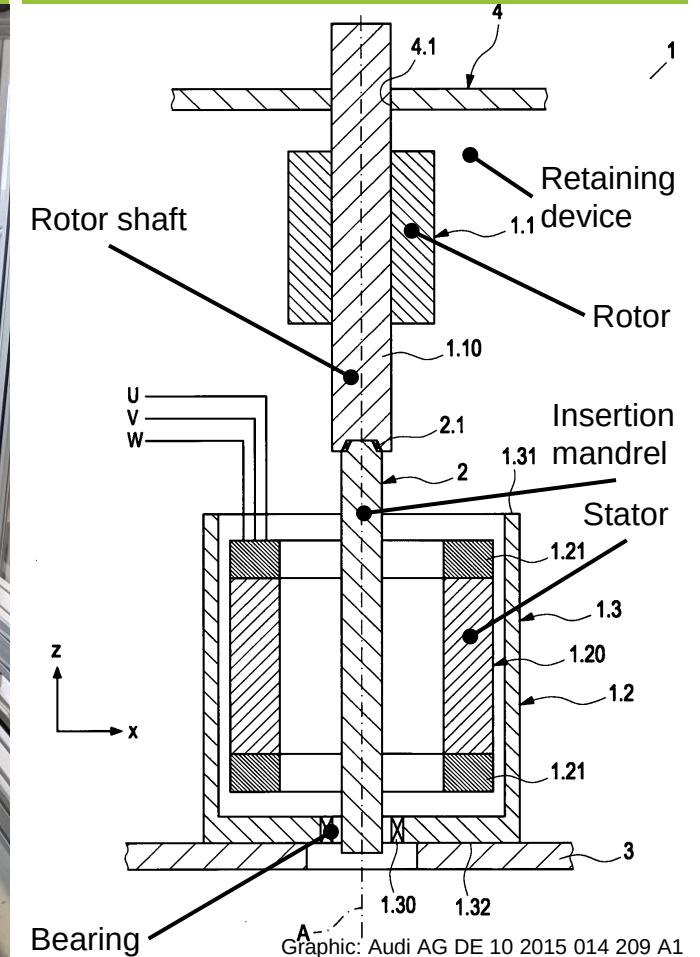


Vertical procedure

Joining the rotor with a solid press frame by massive guiding



Guided rotor joining



The joining of the rotor and stator unit is the central joining operation in the final assembly of an electric motor.

Process

"Marriage" of rotor and stator

Video: Bosch

<https://www.bosch-presse.de/pressportal/de/de/news/video/>



Apart from shrinking the housing onto the stator lamination, a small asynchronous motor can also be installed manually.



Initial components



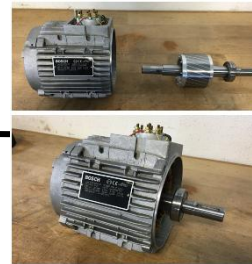
Assembly of the stator in the housing



Bearing shield with fixed bearing



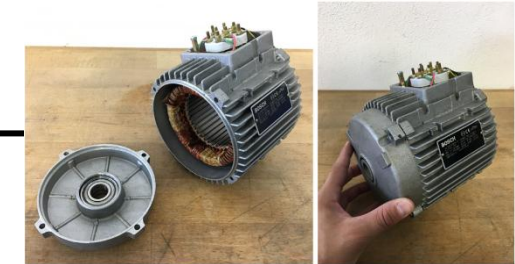
Assembly of the 2nd bearing shield to the housing



Assembly of the rotor shaft



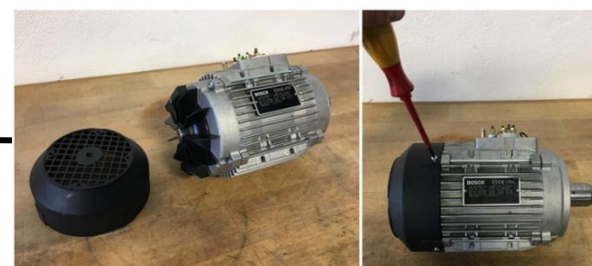
Assembly of the floating bearing



Assembly of the 1st bearing shield



Closing the housing

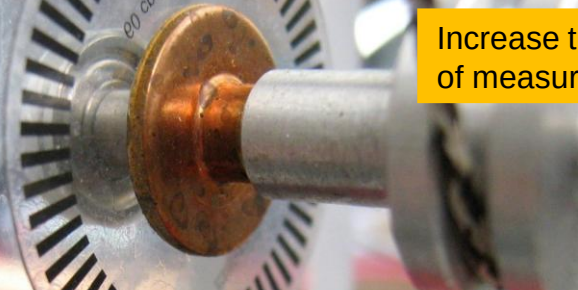
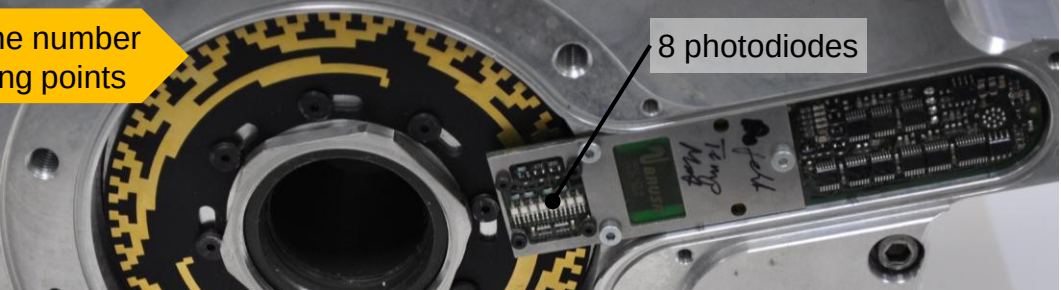
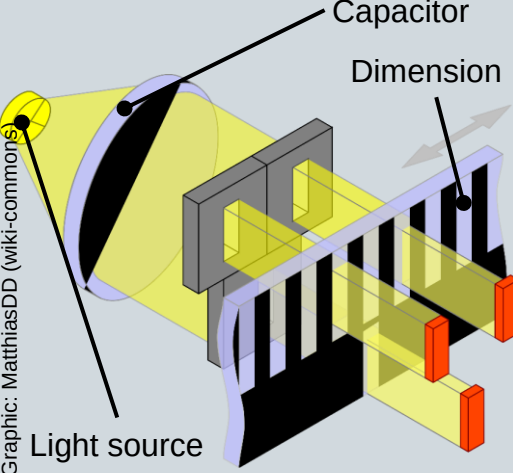


Installing the fan



Handover to testing

By counting pulses, incremental encoders can record the change in position of the electric drive and give this back to monitor precise control tasks.

Incremental encoder	Photodiode - Incremental	Gray code - Absolute
Functionality: - Sensors for detecting changes in position - Scanning measuring principle - Binary code: 0/1 or black/white Variants: - Sliding contacts - Photoelectric - Magnetic	 Graphic: Tycho (wiki-commons) Increase the number of measuring points Differentiation of the series of measuring points	 8 photodiodes 8-row gray code encoder disk (8 bit)
 Capacitor Dimension Light source Graphic: MatthiasDD (wiki-commons)	Features / Advantages - Rotating disk provides alternating illumination of two photodetectors - Receiver determines the change in position of the disk depending on the signals from the two detectors	Features / Advantages - Multi-row encoder disk with binary code - Multi-row read operation in the diode - Only one bit changes with incremental position change - Absolute position detection thus possible
	Disadvantage Exclusive determination of the relative angular change of the measuring disk	Disadvantages - Resolution must match the precision of the value returned - Higher technical complexity

By counting pulses, incremental encoders can record the change in position of the electric drive and give this back to monitor precise control tasks.

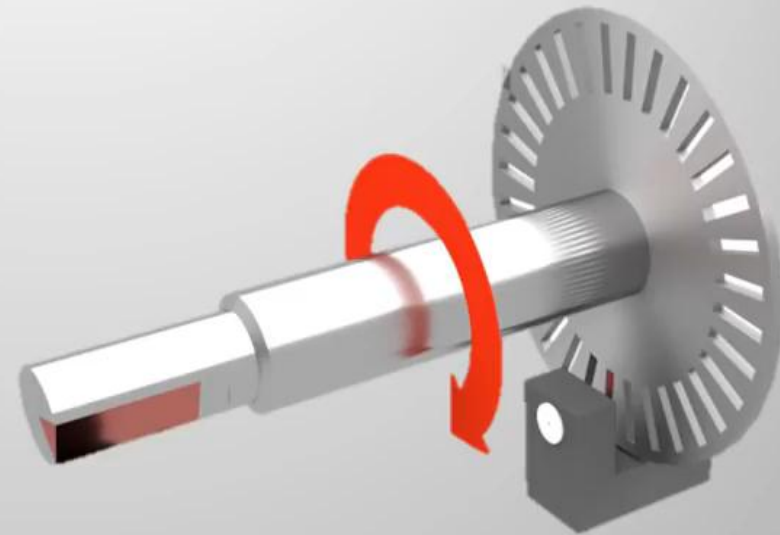
Slide from Lec. 2

Process

Incremental rotor encoder

Video:
Learnchannel

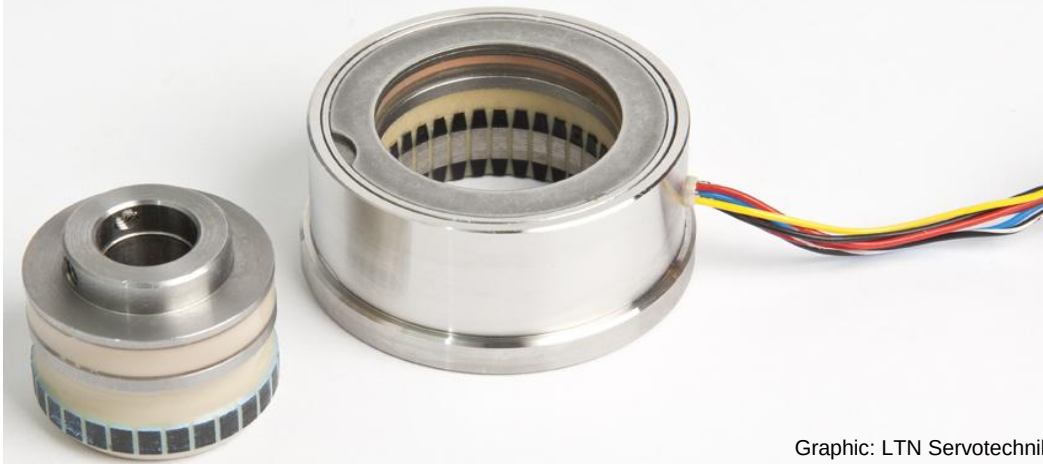
<https://www.youtube.com/watch?v=zzHcsJDV3>



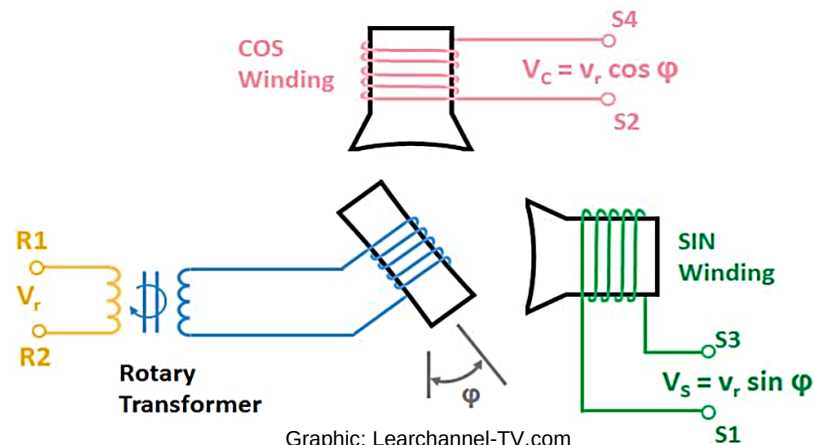
YouTube Learnchannel-TV

Resolvers are angular position sensors that are based on an electromagnetic measuring principle and are widely used due to their high robustness.

The excitation current (R1-R2) is usually transferred inductively (contactless) to the rotor of the resolver

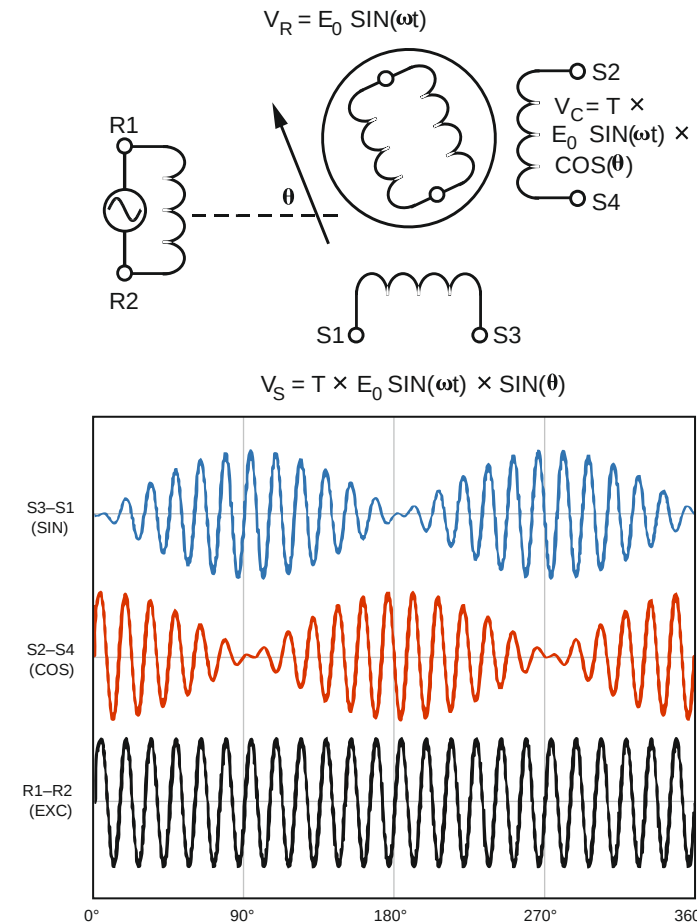


Graphic: LTN Servotechnik



Graphic: Learchannel-TV.com

Resolvers return a signal curve on the sine and cosine windings, from which the rotor position is determined



Graphic: Analog-Devices

Resolvers are angular position sensors that are based on an electromagnetic measuring principle and are widely used due to their high robustness.

Process

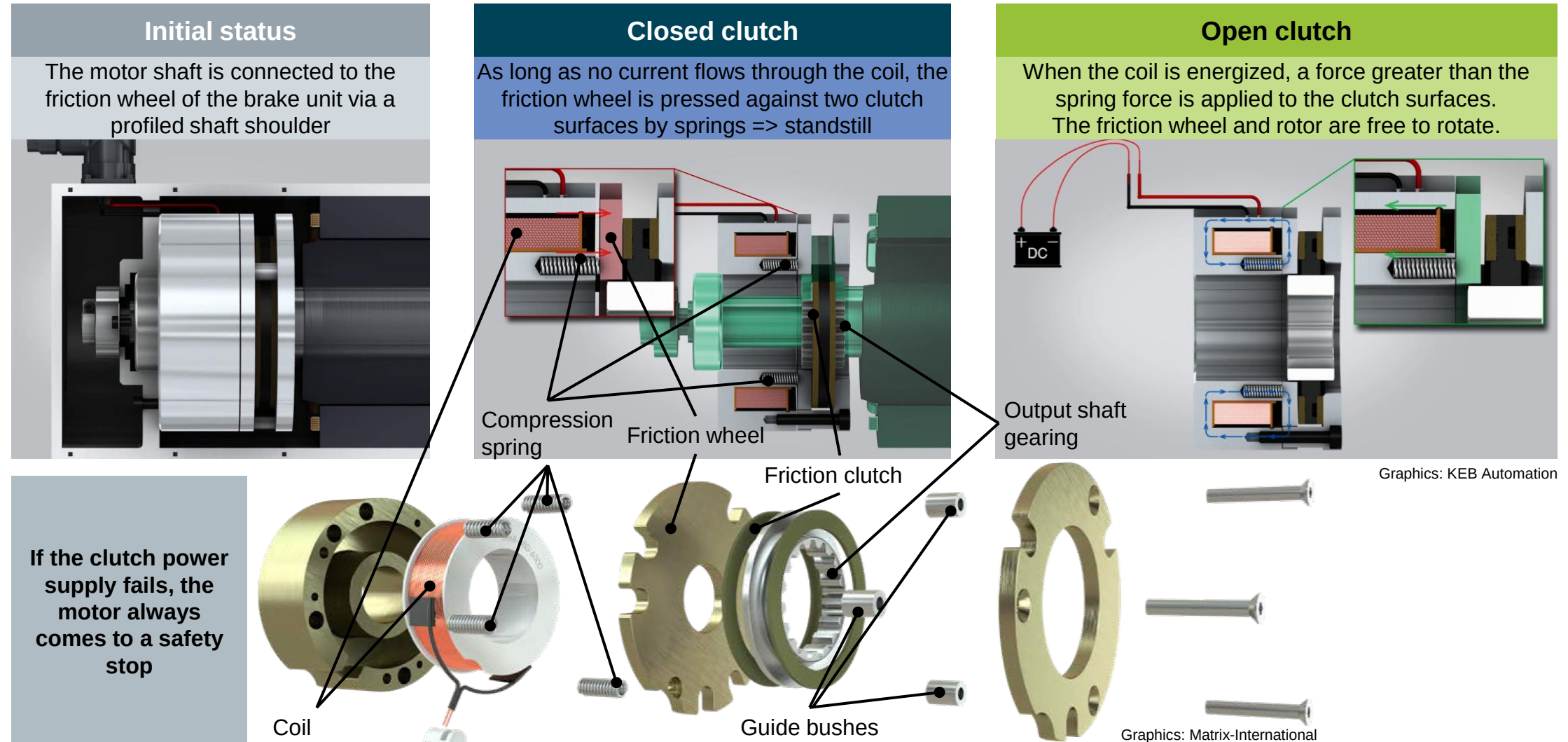
Detailed function of a resolver using
the example of a servo drive

Video: Learnchannel-TV

<https://www.youtube.com/watch?v=x4ZWR-gcEvE&vI=de->



Integrated holding brakes, which function similar to friction clutches and open magnetically, are often installed in servo drives for industrial applications.



The joining of the two main assemblies of an electric motor, the rotor and stator unit, is the central joining operation of the final assembly.

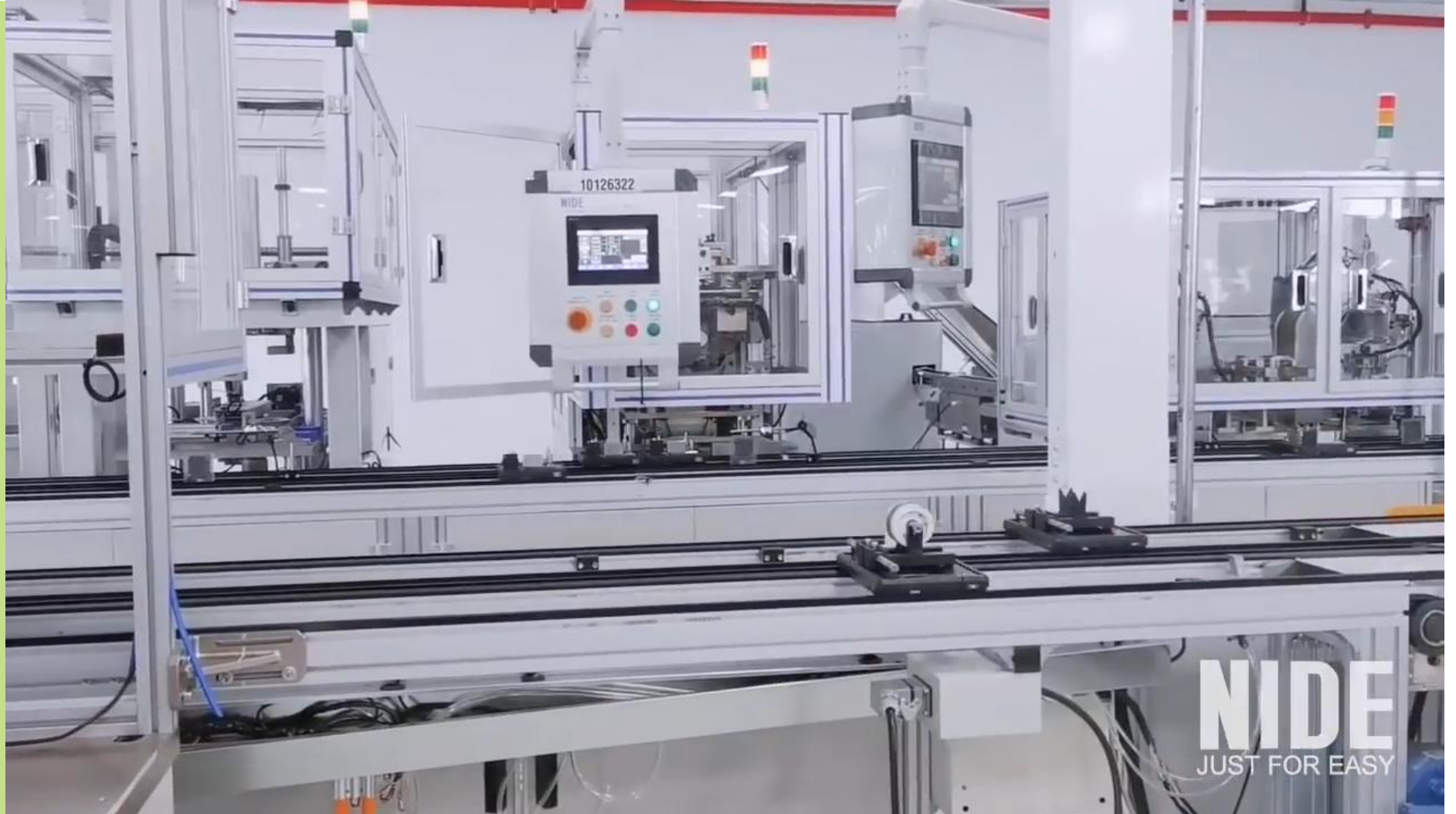
Slide from Lec. 1

Process

Fully automated assembly line

Video: NIDE Group

https://www.youtube.com/watch?v=LAK1yMFfD_w



The contents of the lecture unit can be further deepened with the following specialized literature:

Textbooks

- Tong W. (2014): Mechanical design of electric motors, 1st ed. Boca Raton: CRC Press, Taylor & Francis Group. doi: 10.1201/b16863
- Schneider H. (2013): Auswuchttechnik. Berlin, Heidelberg: Springer Berlin Heidelberg. doi: 10.1007/978-3-642-24914-3
- Schröder D. (2017): Elektrische Antriebe – Grundlagen. Berlin, Heidelberg: Springer Berlin Heidelberg. doi: 10.1007/978-3-642-30096-7
- Böhmer E., Ehrhardt D., Oberschelp W. (2018): Transformatoren und Übertrager. In: Elemente der angewandten Elektronik, Springer Vieweg, Wiesbaden. doi: 10.1007/978-3-8348-9336-9
- Kampker, A. (2014): Elektromobilproduktion. Berlin: Springer Vieweg. doi: 10.1007/978-3-642-42022-1
- Yeadon, W. H.; Yeadon, A. W. (Hg.) (2001): Handbook of small electric motors. New York, NY: McGraw-Hill. ISBN: 978-0070723320
- Tschöke, Helmut; Gutzmer, Peter; Pfund, Thomas (2019): Elektrifizierung des Antriebsstrangs. Berlin, Heidelberg: Springer Berlin Heidelberg. doi: 10.1007/978-3-662-60356-7



The contents of the lecture unit can be further deepened with the following specialized literature:

Dissertations

- Albrecht, T. (2014): Optimierte Fertigungstechnologien für Rotoren getriebeintegrierter PM-Synchronmotoren von Hybridfahrzeugen. Dissertation. Bamberg: Meisenbach. ISBN: 978-3-87525-368-9
- Kühl, A. (2014): Flexible Automatisierung der Statorenmontage mit Hilfe einer universellen ambidexteren Kinematik. Dissertation. Bamberg: Meisenbach. ISBN: 978-3-87525-367-2
- Zerbe, J. (2019): Innovative Wickeltechnologien für Statorspulen zur Erhöhung des Füllfaktors und Reduzierung der Beanspruchungen im Wickelprozess. Dissertation. Berlin: Universitätsverlag der TU Berlin. ISBN: 978-3-7983-3045-0
- Stenzel, P. (2017): Großserientaugliche Nadelwickeltechnik für verteilte Wicklungen im Anwendungsfall der E-Traktionsantriebe. Dissertation. Bamberg: Meisenbach. ISBN: 978-3-87525-425-9
- Dobroschke, Andreas (2011): Flexible Automatisierungslösungen für die Fertigung wickeltechnischer Produkte. Dissertation. Bamberg: Meisenbach. ISBN: 978-3-87525-317-7



The contents of the lecture unit can be further deepened with the following specialized literature:

Articles, patents and other publications

- Gläsel, T.; Bachinski Pinhal, D.; Masuch, M.; Gerling, D.; Franke, J. (2019): Manufacturing Influences on the Motor Performance of Traction Drives with Hairpin Winding. In: 2019 9th International Electric Drives Production Conference (EDPC). 9th International Electric Drives Production Conference (EDPC). Esslingen.
- Gläsel, T.; Masuch, M.; Weigelt, M.; Kühl, A.; Franke, J. (2020): Schlussbericht zu dem Teilvorhaben "Automatisierte Fertigungstechnologien zum Biegen, Montieren und Schalten von innovativen Formspulenwicklungen" im Verbundprojekt "PRO-E-Traktion" ("Automatisierte und robuste Produktionssysteme für E-Traktionsantriebe").
- Dr. Risch, F.; Schilp, H.; Schlag, P.; Dr. Beckmöller, S.; Kehl, N.; Halder, H. et al. (2016): Hochflexible Produktionssysteme für effizienzgesteigerte E-Traktionsantriebe (HeP-E). Abschlussbericht des BMBF-Verbundprojekts. Hg. v. Verband Deutscher Maschinen- und Anlagenbau e.V. Forum Glastechnik. Frankfurt am Main.
- Fleischer, J.; Haag, S.; Hofmann, J. (2017): Quo Vadis Winding Technology? A study on the state of the art and research on future trends in automotive engineering. Hg. v. J. Fleischer. Institute of Production Science (wbk). Karlsruhe.
- Henke, M.; Narjes, G.; Hoffmann, J.; Wohlers, C.; Urbanek, S.; Heister, C. et al. (2018): Challenges and Opportunities of Very Light High-Performance Electric Drives for Aviation. In: *Energies* 11 (2), S. 344. DOI: 10.3390/en11020344.
- Kühl, A.; Riedel, A.; Vogel, A.; Hartl, S.; Gläsel, T.; Masuch, M.; Franke, J.: Robot-based Production of Electric Motors with Hairpin Winding Technology. In: Proceedings of the World Congress on Engineering and Computer Science 2019 (WCECS), 257-256.
- Schmidt, J. (2015): Verfahren zur Montage einer einen Rotor und einen Stator umfassenden Drehfeldmaschine. Patentanmeldung durch AUDI AG. Ingolstadt





FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

**Institute for Factory Automation
and Production Systems**



Friedrich-Alexander-Universität
Technische Fakultät

THANK YOU